Cosmology: From the Hubble Law to the Early Universe

MAIS Physics & Astronomy Club (PAC) · Olympiad lecture + problem handout

15 lecture sections · 30 problems · 5 interactive experiments · 6 vector figures

Begin with a question a ruler could answer: *what changes when the universe expands?* From that picture we build redshift, gravity, cosmic distances and the thermal history of the universe. Sophisticated ideas are introduced through physical reasoning before the formalism. The 30 problems include website-sourced adaptations and new IOAA-inspired challenges; every one has a worked solution.

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1. What cosmology actually asks

Astronomy often asks, *What is that object?* Cosmology asks a larger question: **what rules describe the universe as a whole?** A star has a surface, a centre and surroundings. The universe need not have any of these. We cannot step outside it, look at its edge, or repeat the experiment with a second universe. Instead, we collect light from different places and different past times, then ask which simple model explains all of those observations together.

The surprising discovery is that very large scales can be simpler than small ones. A glass of water contains an enormous number of individually complicated molecules, yet a single density describes it very well. Likewise, galaxies are complicated, but averaging over sufficiently large volumes lets us describe the cosmic background by a density, a pressure and an expansion rate. Structure has not disappeared; we have chosen which questions to ask.

> **The working picture.** On large scales the universe is approximately homogeneous and isotropic, its spatial geometry is close to flat, and its scale factor is increasing. The hot early universe left light nuclei and a microwave background. Small initial density variations grew into galaxies and clusters. A matter component that gravitates but is not ordinary luminous material, and a late-time accelerating component consistent with a cosmological constant, complete the minimal **ΛCDM** picture.

Homogeneous means "the same from place to place" at a specified cosmic time. **Isotropic** means "the same in every direction" about a point. These are not synonyms. A uniform magnetic field can be homogeneous without being isotropic: it selects a direction. A spherically symmetric distribution centred on us could look isotropic to us without being homogeneous. Isotropy about every position implies homogeneity. Our near-isotropic microwave sky, together with galaxy surveys and the Copernican assumption that we do not occupy a special central position, supports the large-scale model.

The **cosmological principle** is an approximation, not a claim that your room looks the same as a galaxy cluster. Homogeneity emerges only after substantial averaging; the precise scale depends on the statistic being measured. Cosmic time is the proper time measured by observers moving with the mean cosmic matter, called **comoving observers**. "The universe now" means a surface of equal cosmic time in this model, not everything that our telescope can see at once.

Three different kinds of evidence

1. **Geometry and expansion:** redshifts, standardizable candles, standard rulers and gravitational-lens time delays test the relation between distance, redshift and cosmic time.
2. **Contents and history:** light-element abundances, the cosmic microwave background (CMB), and the growth of structure test the hot early universe and its ingredients.
3. **Consistency:** the same parameters must fit independent observations. A model that explains one graph but fails another is not yet a successful cosmology.

A measurement is not the same thing as an interpretation. We measure wavelengths, angles, fluxes, count rates and temperatures. "Dark energy," "the age of the universe," and "the curvature of space" are inferred within a framework. Good Olympiad solutions keep that chain of inference visible.

> ***Scope and notation.*** This handout goes beyond the minimum IOAA syllabus. The core route uses algebra, geometry and physical reasoning. Sections labelled **Extension** introduce calculus, the FLRW metric and early-universe theory; they are enrichment, not a claim that tensor calculus is required in a standard IOAA solution. Throughout, $a_0 = 1$, z is cosmological redshift unless stated otherwise, ρ is mass density, $u = \rho c^2$ is energy density, and P is pressure. An overdot means d/dt.

Constants and a deliberately simple benchmark

Quantity	Value used here	Meaning
c	$299\,792.458\ \mathrm{km\,s^{-1}}$	speed of light
G	$6.6743 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$	gravitational constant
$1\ \mathrm{Mpc}$	$3.0856776 \times 10^{19}\ \mathrm{km}$	megaparsec
$1\ \mathrm{Gyr}$	$3.15576 \times 10^{16}\ \mathrm{s}$	Julian gigayear
σ_{SB}	$5.670374419 \times 10^{-8}\ \mathrm{W\,m^{-2}\,K^{-4}}$	Stefan–Boltzmann constant
b_{W}	$2.89777 \times 10^{-3}\ \mathrm{m\,K}$	Wien constant for the peak of B_λ
k_{B}	$1.380649 \times 10^{-23}\ \mathrm{J\,K^{-1}}$	Boltzmann constant
$T_{\mathrm{CMB},0}$	$2.725\ \mathrm{K}$	rounded present CMB temperature
H_0	$70\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$	teaching benchmark, not a consensus measurement
$\Omega_{m0}, \Omega_{\Lambda 0}$	$0.3,\ 0.7$	rounded flat benchmark

The benchmark sets radiation and curvature to zero **only when studying late-time expansion**. That approximation must be switched off when discussing the early universe or matter–radiation equality. For those calculations we supply a radiation density explicitly. Parameters taken from an individual past-paper question override this table. Modern precision cosmology quotes uncertainties and correlations; these rounded numbers are for transparent calculations, not a fit to the latest data.

2. Expansion without a centre

Imagine marks on a rubber sheet. Increase every separation by the same factor, and each mark sees the other marks recede. No mark is the centre of the stretching. This is a useful analogy for a mathematical relation, not a claim that space is made of rubber, that the universe expands into a room, or that a two-dimensional sheet is the whole universe.

Let $\boldsymbol{\chi}$ be a fixed **comoving coordinate** and $a(t)$ the dimensionless **scale factor**. In flat space the proper separation of two comoving galaxies is

$$D(t) = a(t)\,\chi, \qquad H(t) \equiv \frac{\dot{a}}{a}, \qquad \dot{D} = HD. \tag{1}$$

The last equality applies to the separation of ideal comoving galaxies, not to the size of a ruler or an atom. The coordinate χ is the separation the pair would have today because $a_0 = 1$. The quantity H is a fractional growth rate. If $H = 70\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, a separation of one megaparsec grows by 70 kilometres in a second, while two megaparsecs grow by 140 kilometres in that second. The units are those of inverse time.

$$H_0^{-1} = 13.97\ \text{Gyr}, \qquad \frac{c}{H_0} = 4.283\ \text{Gpc} = 13.97\ \text{Gly} \quad (H_0 = 70). \tag{2}$$

The **Hubble time** H_0^{-1} is a timescale, not automatically the age. The age depends on the entire history $H(t)$. The **Hubble radius** c/H is a local expansion length, not automatically the edge of the observable universe.

INTERACTIVE EXPERIMENT 1

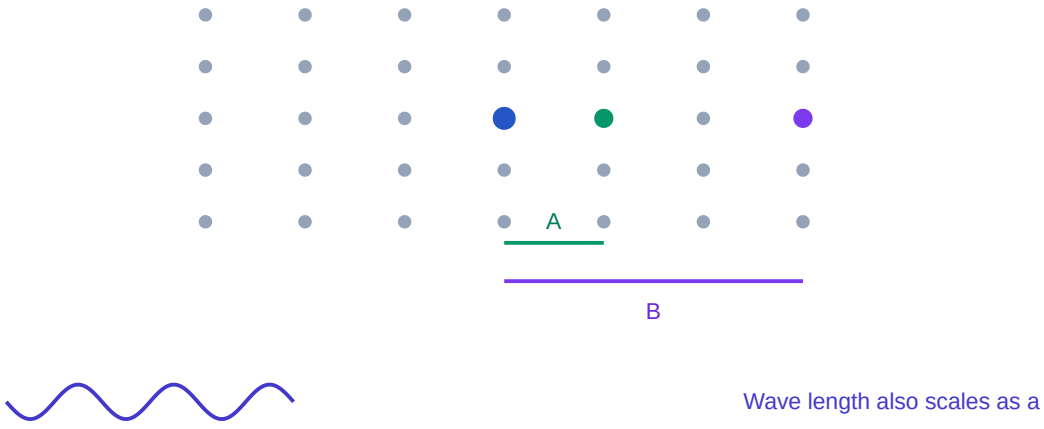
Every separation grows. No galaxy is the centre.

The dots are ideal comoving galaxies. Their plotted sizes stay fixed while their separations scale. Galaxy A has χ = 100 Mpc; B has χ = 300 Mpc. This is a kinematic illustration, not a solution for H(t).

Scale factor	Galaxy A: 100 Mpc × a	Galaxy B: 300 Mpc × a
1.00	**100 Mpc**	**300 Mpc**

Distance ratio B/A
3.00

Blue: observer · Green: A · Purple: B



At a = 1.00, the separations are 100 and 300 Mpc. Their ratio stays 3; the galaxy symbols do not expand.

Try this. Double *a*. Do the galaxies get larger? Does B/A change? Move the conceptual observer to another dot: the same proportional stretching still applies.

This is a snapshot of the interactive experiment. Open the HTML edition to change its controls.

What expands, and what does not?

Freely moving, widely separated galaxies participate in the mean Hubble flow. Galaxies, stars, planets and atoms are not miniature copies of that flow. They are bound or otherwise locally governed systems. The Earth does not double in size when the cosmic scale factor doubles. Even the separation of two galaxies in a bound group need not follow $D = a\chi$.

A real galaxy's proper distance has a background part and a **peculiar velocity** part:

$$\dot{D} = HD + v_{\mathrm{pec}}. \tag{3}$$

Peculiar velocities arise from local gravitational interactions. A few hundred kilometres per second can overwhelm the Hubble velocity of a nearby galaxy. Andromeda is blueshifted: its motion toward us is not a failure of cosmology, because it is not an isolated comoving test particle in the smooth Hubble flow.

Example 1 — changing the observer does not create a centre. Suppose galaxies have positions $\mathbf{r}_i = a\boldsymbol{\chi}_i$. Relative to galaxy j,

$$\mathbf{r}_i - \mathbf{r}_j = a(\boldsymbol{\chi}_i - \boldsymbol{\chi}_j), \qquad \dot{\mathbf{r}}_i - \dot{\mathbf{r}}_j = H(\mathbf{r}_i - \mathbf{r}_j). \tag{4}$$

Every comoving galaxy sees the same linear law. The origin was a coordinate choice. **Check:** setting $H = 0$ removes the Hubble flow without moving any coordinate labels.

A first map of cosmic history

The ordering is much more important than memorizing a dozen uncertain decimal places. The diagram is a timeline, **not a size-to-scale drawing**.

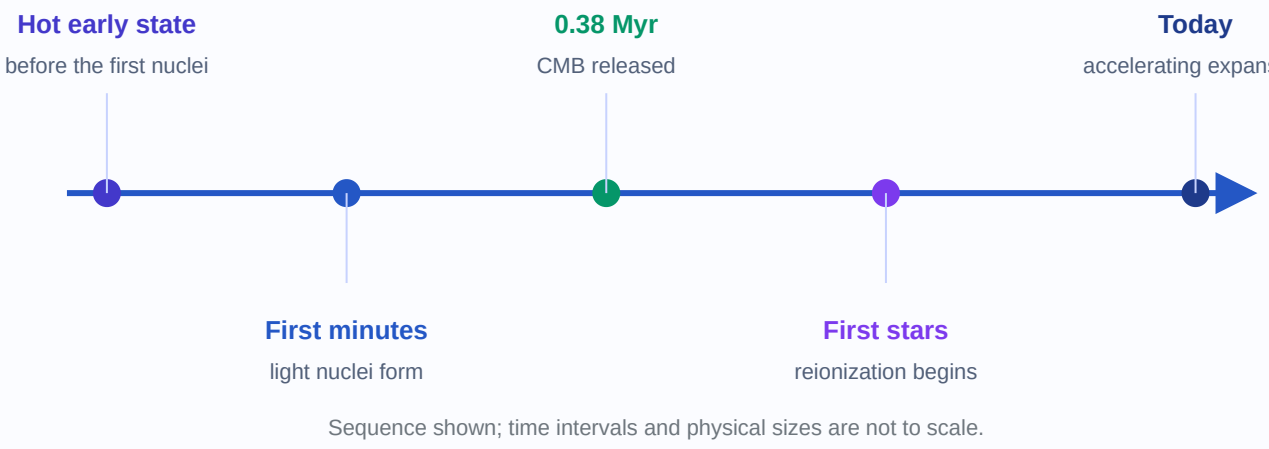


Figure 1. A thermal history, not a map of an explosion. The CMB is released long after the light elements form.

“Big Bang” names the hot, dense early state and the subsequent expansion. Extrapolating classical equations all the way to $a = 0$ produces a singularity, but that extrapolation exceeds the tested theory. It is not a demonstrated description of a literal first instant, and it is not an explosion at a particular location in pre-existing space.

3. Redshift: a clock carried by light

Define redshift using a rest-frame spectral feature:

$$1 + z \equiv \frac{\lambda_{\rm obs}}{\lambda_{\rm em}} = \frac{\nu_{\rm em}}{\nu_{\rm obs}}. \tag{5}$$

For light propagating through the smooth expanding background,

$$\boxed{1 + z = \frac{a(t_{\rm obs})}{a(t_{\rm em})}}, \qquad a_{\rm em} = \frac{1}{1+z} \quad (a_{\rm obs} = 1). \tag{6}$$

Here is the idea without machinery. Send two successive wave crests. Both follow the same kind of light path, but the second starts a little later and arrives a little later. In an expanding universe the interval between arrivals is stretched by the same ratio as the scale factor. A wave's period is a clock; its wavelength is that clock multiplied by c.

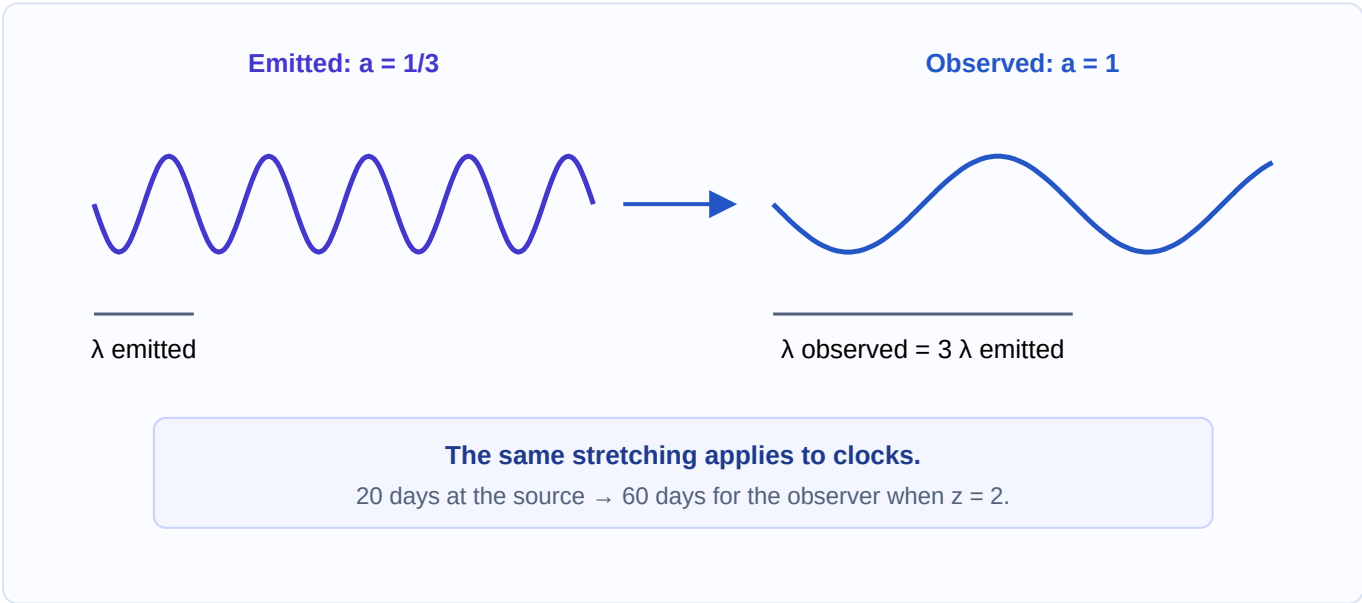


Figure 2. Redshift stretches wavelength and duration by the same factor. The wave amplitudes are schematic and are not an intensity calculation.

Extension: a short derivation. For radial light in the metric of Section 4, $d\chi = c\,dt/a(t)$. The comoving separation of source and observer is fixed, so the integral for the first crest equals the integral for the next. For small crest intervals, $\delta t_{\rm obs}/a_{\rm obs} = \delta t_{\rm em}/a_{\rm em}$. Hence periods, wavelengths and observed durations all acquire the factor $a_{\rm obs}/a_{\rm em}$.

A supernova at $z = 1$ therefore has its light-curve timescales stretched by two. This **cosmological time dilation** is an observationally important consequence, not an optional decoration of the redshift formula.

Redshift is not one unique kind of velocity

For a purely radial special-relativistic Doppler shift in flat spacetime,

$$1 + z_{\mathrm{Dop}} = \sqrt{\frac{1+\beta}{1-\beta}}, \qquad \beta = \frac{v}{c}, \qquad \frac{v}{c} = \frac{(1+z)^2 - 1}{(1+z)^2 + 1}. \tag{7}$$

At small z, $v \simeq cz$. At large cosmological redshift, inserting cz into this formula and calling the result “the galaxy's velocity” is generally wrong. Cosmological redshift compares the scale factor at two events. A recession rate HD compares proper distances on a chosen cosmic-time slice. They are different constructions. In curved spacetime there is no unique, distance-independent way to compare remote velocity vectors.

Observed shifts multiply rather than simply add:

$$1 + z_{\mathrm{obs}} = (1 + z_{\mathrm{cos}})(1 + z_{\mathrm{pec}})(1 + z_{\mathrm{grav}}) \cdots . \tag{8}$$

For small shifts the fractional changes approximately add, which explains the familiar low-redshift formula. A careful high-redshift analysis also accounts for the observer's motion, the source's environment and the measurement band.

A useful trap. A source at $z = 2$ is **not** moving through its local neighbourhood at $2c$. Its light is emitted locally at c, and every local observer measures light at c. The scale factor at emission was $1/3$. Its proper distance then, its proper distance now, and its light-travel time are three different quantities.

4. Geometry: measuring a universe from inside

Extension — the FLRW metric

A metric is a rule for turning coordinate differences into measured distances and times. You already use one on a map: a centimetre on the map corresponds to a particular distance on the ground. In cosmology the conversion factor changes with time.

For a homogeneous and isotropic universe, the Friedmann–Lemaître–Robertson–Walker (FLRW) line element can be written

$$ds^2 = -c^2 dt^2 + a^2(t)\left[d\chi^2 + S_k^2(\chi)\left(d\theta^2 + \sin^2\theta\, d\phi^2\right)\right], \tag{9}$$

$$S_k(\chi) = \begin{cases} R_c \sin(\chi/R_c), & k = +1, \\ \chi, & k = 0, \\ R_c \sinh(\chi/R_c), & k = -1. \end{cases} \tag{10}$$

Here χ is a radial **geodesic coordinate distance** in the comoving spatial geometry and R_c is its curvature radius today. Some books instead use a dimensionless χ and put dimensions into a: the physics is identical, but mixing conventions causes errors. This handout keeps $a_0 = 1$ and gives χ and R_c units of length. With this convention the curvature constant k has units of inverse length squared: $k = +1/R_c^2$, 0, or $-1/R_c^2$. Only its sign is the usual label $+1, 0, -1$.

- $ds^2 < 0$: a timelike separation; a massive observer can experience it, with $d\tau^2 = -ds^2/c^2$.
- $ds^2 = 0$: light propagation.
- On a $dt = 0$ slice, ds is a proper spatial distance.
- At fixed t and χ, a small transverse angle $d\theta$ corresponds to proper length $aS_k(\chi)d\theta$.

For $k = +1$, the spatial geometry is locally like a three-dimensional sphere. Its finite volume has no boundary, just as the two-dimensional surface of a ball has no edge. For $k = 0$ it is locally Euclidean. For $k = -1$ it is locally hyperbolic. **Geometry does not by itself settle topology:** a flat space can have identifications, and a negatively curved space need not be infinite. Do not infer the entire universe's shape from the sign of k alone.

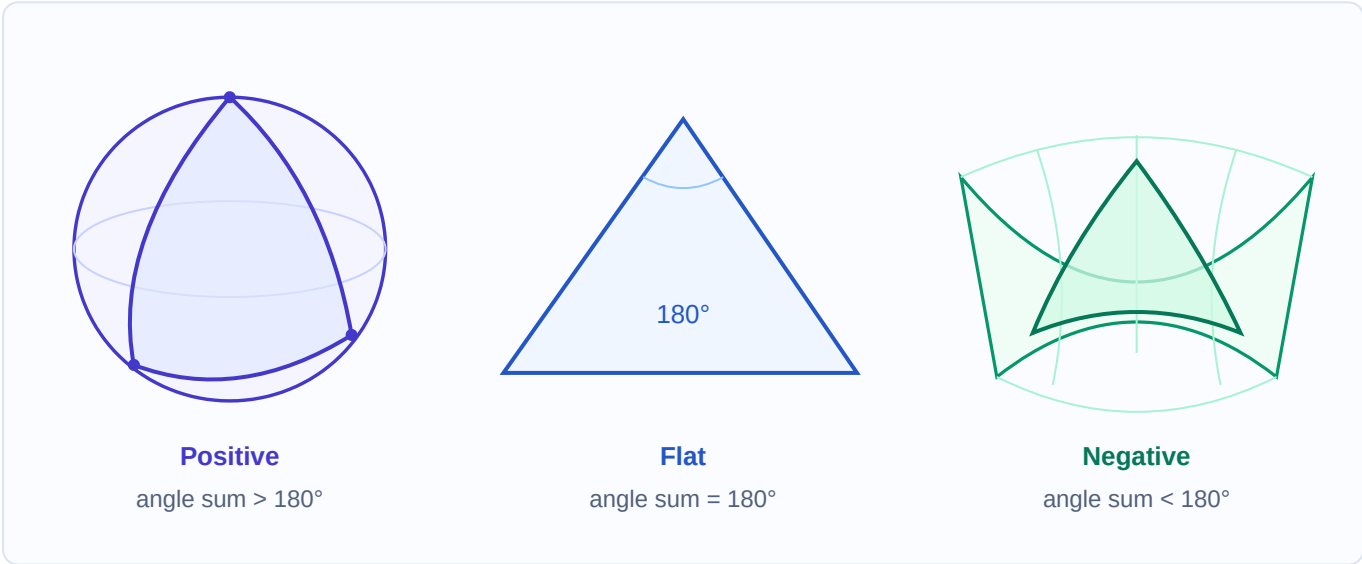


Figure 3. Curvature is measured by relations inside space. These are local geometry sketches, not claims about the global topology of the universe.

On a sphere of radius R_c, a circle drawn at geodesic radius r has circumference

$$C(r) = 2\pi R_c \sin(r/R_c) = 2\pi r \left[1 - \frac{r^2}{6R_c^2} + O\left(\frac{r^4}{R_c^4}\right)\right]. \tag{11}$$

A small enough patch looks flat. This is why local measurements can be exquisitely Euclidean even if the curvature radius is finite. The relevant small parameter is r/R_c, not r measured in metres.

Curvature is not the centre of an explosion

The curvature refers to relations among distances and angles **inside space**. It does not require a physical fourth spatial direction into which our universe bends. Embedding pictures can help intuition, but the intrinsic measurements—circumferences, triangle angles and volumes—are what matter.

5. Why the expansion changes

A Newtonian route to the Friedmann equation

First imagine a uniform sphere of pressureless matter. Follow a test galaxy at its edge, at physical radius $R = a\chi$. The enclosed mass $M = (4\pi/3)\rho R^3$ stays fixed as the sphere expands. Newton's shell theorem lets us treat that mass as if it were concentrated at the centre. The test galaxy's energy per unit mass is

$$\frac{1}{2}\dot{R}^2 - \frac{GM}{R} = E. \tag{12}$$

Divide by $R^2/2$ and use $R = a\chi$:

$$H^2 = \frac{8\pi G}{3}\rho + \frac{2E}{\chi^2 a^2}. \tag{13}$$

Identifying the last term with spatial curvature gives the first Friedmann equation. This argument builds intuition; the full result, including pressure and the meaning of the metric, belongs to general relativity.

$$\boxed{H^2 = \frac{8\pi G}{3}\rho - \frac{kc^2}{a^2} + \frac{\Lambda c^2}{3}}. \tag{14}$$

With the length-valued comoving coordinates above, a is dimensionless and the signed k carries units of inverse length squared. Other books make k dimensionless by a different coordinate choice. To avoid convention errors in calculations, **use the density-parameter form below**.

Define the critical density and density parameters by

$$\rho_{\mathrm{crit}}(t) = \frac{3H^2}{8\pi G}, \qquad \Omega_i = \frac{\rho_i}{\rho_{\mathrm{crit}}}, \qquad \Omega_k = -\frac{kc^2}{a^2H^2}, \qquad \Omega_\Lambda = \frac{\Lambda c^2}{3H^2}. \tag{15}$$

Then, exactly within FLRW cosmology,

$$1 = \Omega_m + \Omega_r + \Omega_\Lambda + \Omega_k. \tag{16}$$

For the normalized scale factor $a_0 = 1$,

$$\boxed{E^2(z) \equiv \frac{H^2(z)}{H_0^2} = \Omega_{r0}(1+z)^4 + \Omega_{m0}(1+z)^3 + \Omega_{k0}(1+z)^2 + \Omega_{\Lambda 0}}. \tag{17}$$

For a more general non-interacting component with constant equation-of-state parameter $w_i = P_i/u_i$, replace its contribution by $\Omega_{i0}(1+z)^{3(1+w_i)}$.

Example 2 — the density that goes with the benchmark. Convert H_0 before using SI constants:

$$H_0 = \frac{70}{3.0856776 \times 10^{19}}\ \mathrm{s}^{-1} = 2.26855 \times 10^{-18}\ \mathrm{s}^{-1}, \tag{18}$$

$$\rho_{\mathrm{crit},0} = 9.20 \times 10^{-27}\ \mathrm{kg\,m^{-3}}, \qquad u_{\mathrm{crit},0} = 8.27 \times 10^{-10}\ \mathrm{J\,m^{-3}}. \tag{19}$$

That is only a few hydrogen masses per cubic metre. “Critical” does not mean dense in everyday terms. It marks the density needed for spatial flatness at a given expansion rate, after all energy contributions are included.

Conservation: the first law applied to expansion

Take a comoving volume $V \propto a^3$. For a uniform fluid with no matter flowing across its boundary and no heat exchange between separate components, $d(uV) = -P\,dV$. This gives

$$\boxed{\dot{u} + 3H(u + P) = 0}. \tag{20}$$

For $P = wu$ with constant w,

$$u \propto a^{-3(1+w)}, \qquad \rho_m \propto a^{-3}, \qquad u_r \propto a^{-4}, \qquad u_\Lambda = \text{constant}. \tag{21}$$

Matter dilution is a counting argument: the same particles occupy a larger volume. Radiation loses photon energy as well: number density falls as a^{-3} and each photon's energy falls as a^{-1}. A cosmological constant has $w = -1$, so its energy density remains constant as volume grows. This does **not** mean that a small box is an unlimited source of extractable useful energy; global gravitational energy is not bookkept like the energy of a gas cylinder in a laboratory.

Pressure gravitates too

The acceleration equation is

$$\boxed{\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3P}{c^2}\right) + \frac{\Lambda c^2}{3}}. \tag{22}$$

If Λ is included as a fluid with $u_\Lambda = \rho_\Lambda c^2$ and $P_\Lambda = -u_\Lambda$, omit the separate Λ term and include that fluid in the sum. **Choose one bookkeeping convention.** Combining the two equations with local conservation shows that they are not three unrelated assumptions.

For a component to accelerate expansion by itself, it needs

$$u + 3P < 0 \quad \Longleftrightarrow \quad w < -\frac{1}{3}. \tag{23}$$

Radiation has $w = 1/3$ and decelerates expansion more strongly, at fixed energy density, than pressureless matter. Negative pressure is not a gas "sucking from outside." It is a term in the stress-energy that tells gravity how that component behaves. Uniform pressure has no pressure-gradient force; its gravitational effect remains.

Define the deceleration parameter

$$q \equiv -\frac{a\ddot{a}}{\dot{a}^2} = -1 - \frac{\dot{H}}{H^2}. \tag{24}$$

Positive q means slowing expansion; negative q means accelerating expansion. An expanding universe can have a decreasing H while $\dot{D} = a\chi H$ increases. Do not confuse $\dot{H} < 0$ with $\ddot{a} < 0$.

6. Solving the expansion history

The first Friedmann equation is an energy-like equation. Once the contents are specified, it determines how rapidly a changes. The useful conversion is

$$dt = \frac{da}{aH(a)} = -\frac{dz}{(1+z)H(z)}. \tag{25}$$

Single-component universes

For a flat universe dominated by one constant-w component, with $w > -1$,

$$a(t) \propto t^{\frac{2}{3(1+w)}}, \qquad H = \frac{2}{3(1+w)t}. \tag{26}$$

Dominant component	w	$a(t)$	Age–rate relation
radiation	$1/3$	$t^{1/2}$	$t = 1/(2H)$
pressureless matter	0	$t^{2/3}$	$t = 2/(3H)$
cosmological constant	-1	e^{Ht}	no finite Big Bang age within this pure model
curvature-dominated toy model	effective $-1/3$	t	$t = 1/H$

The last row is not an ordinary fluid dominating a flat universe; it describes the behaviour of an empty, negatively curved Milne model. These exact models are benchmarks. The real universe passes through

different regimes and cannot be assigned one power law for its entire history.

Example 3 — why Einstein–de Sitter is not the benchmark age. In a flat, matter-only model, $H = H_0 a^{-3/2}$. Thus

$$t_0 = \frac{1}{H_0}\int_0^1 \frac{da}{a\,a^{-3/2}} = \frac{1}{H_0}\int_0^1 a^{-1/2}da = \frac{2}{3H_0}. \tag{27}$$

For $H_0 = 70$, this is 9.31 Gyr, not 13.97 Gyr. The missing ingredient is the expansion history, not a numerical conversion error.

The flat matter-plus-Λ benchmark

Neglect radiation at late times. With $\Omega_{m0} + \Omega_{\Lambda 0} = 1$ and positive densities,

$$a(t) = \left(\frac{\Omega_{m0}}{\Omega_{\Lambda 0}}\right)^{1/3} \sinh^{2/3}\left(\frac{3}{2}H_0\sqrt{\Omega_{\Lambda 0}}\,t\right), \tag{28}$$

$$\boxed{t_0 = \frac{2}{3H_0\sqrt{\Omega_{\Lambda 0}}}\operatorname{arsinh}\sqrt{\frac{\Omega_{\Lambda 0}}{\Omega_{m0}}}}. \tag{29}$$

For $(0.3, 0.7, 70)$ this gives approximately 13.47 Gyr. The expression is a useful check on numerical integration, but it is not valid through the radiation era without modification.

The lookback time to redshift z and the cosmic age at emission are

$$t_{\rm LB}(z) = \frac{1}{H_0}\int_0^z \frac{dz'}{(1+z')E(z')}, \qquad t(z) = t_0 - t_{\rm LB}(z). \tag{30}$$

Equality is not acceleration

Matter and Λ have equal densities when

$$1 + z_{m\Lambda} = \left(\frac{\Omega_{\Lambda 0}}{\Omega_{m0}}\right)^{1/3}. \tag{31}$$

Acceleration starts when the active gravitational density vanishes: $\rho_m + \rho_\Lambda + 3P_\Lambda/c^2 = \rho_m - 2\rho_\Lambda = 0$. Thus

$$\boxed{1 + z_{\rm acc} = \left(\frac{2\Omega_{\Lambda 0}}{\Omega_{m0}}\right)^{1/3}}. \tag{32}$$

For the benchmark, $z_{m\Lambda} \simeq 0.327$ and $z_{\mathrm{acc}} \simeq 0.672$. Acceleration begins earlier, at a higher redshift, because Λ acquires dynamical importance before its density actually exceeds the matter density. **Never use "domination" as a synonym for "the onset of acceleration."**

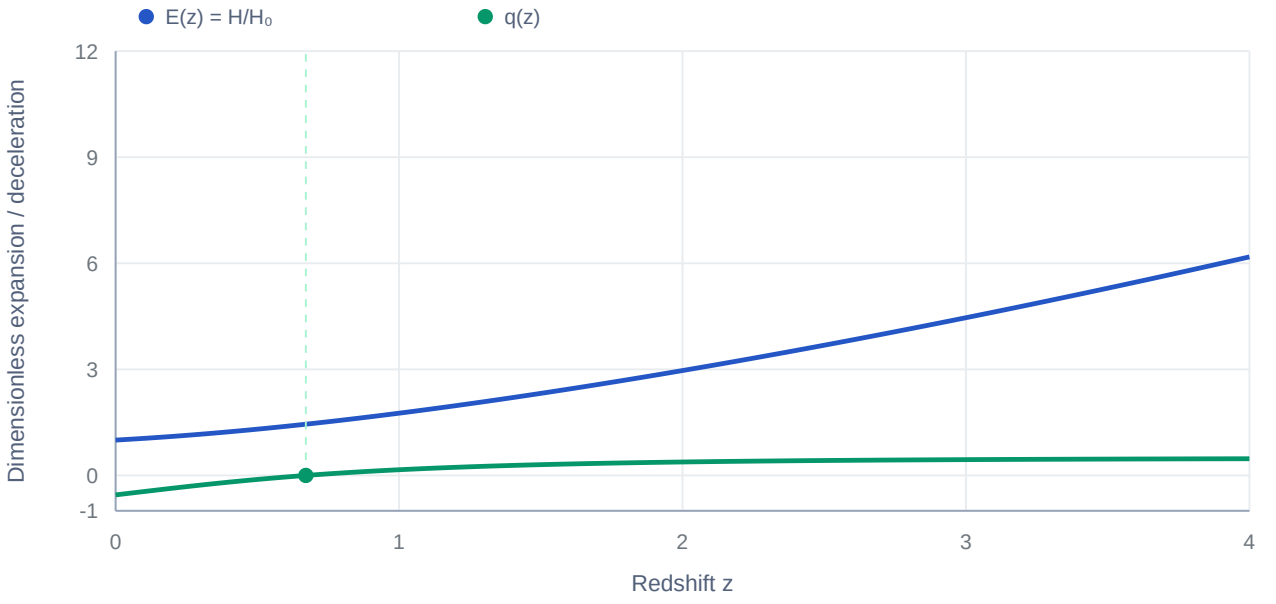
INTERACTIVE EXPERIMENT 2

Change the contents. Change the cosmic clock.

A flat matter-plus-Λ model, with radiation neglected. Blue shows E(z) = H/H₀; green shows q(z). Negative q means accelerating expansion. The age integral includes the supplied matter and Λ components only.

ΩΛ0 = 1 − Ωm0	Present age	q₀
0.70	**13.47 Gyr**	**-0.55**

Matter–Λ equality z	Acceleration begins at z
0.326	**0.671**



Ωm0 = 0.30, ΩΛ0 = 0.70. The age is 13.47 Gyr and q₀ = -0.55. Radiation is intentionally omitted.

Try this. Set $\Omega_{m0} = 1$. The age becomes $2/(3H_0)$. Return to 0.3: find the redshift where q crosses zero, and compare it with the lower redshift where the two densities are equal.

This is a snapshot of the interactive experiment. Open the HTML edition to change its controls.

Curvature and cosmic fate

Spatial geometry and future expansion were once treated as almost the same question. With Λ they are not. A positively curved universe can expand forever if Λ is large enough; a flat universe containing only matter decelerates forever. To decide the fate, examine whether $H^2(a)$ can reach zero at a future $a > 1$, and then use the acceleration equation. The sign of curvature alone is insufficient.

A static solution requires both $H = 0$ and $\ddot{a} = 0$. The Einstein static universe balances matter, positive curvature and Λ, but the balance is unstable: a small expansion lowers the matter density and lets Λ win; a small contraction does the reverse. This is a valuable limiting-case problem, not the model we observe.

7. Five distances, five different questions

At large redshift, “How far away is it?” is incomplete. Ask **which operation defines the distance?** The following definitions reduce to the same familiar distance nearby but differ substantially far away.

Distance	Operational question	Flat-universe relation
lookback distance ct_{LB}	how long has the light travelled?	integrate dt
comoving radial distance D_C	what would a present-day ruler measure?	$c\int_0^z dz'/H(z')$
proper distance then D_{em}	how far was it when the light left?	$D_C/(1+z)$
angular-diameter distance d_A	how large does a known ruler look?	$D_C/(1+z)$
luminosity distance d_L	how faint does a known luminosity appear?	$(1+z)D_C$

In a curved universe the transverse comoving distance is $D_M = S_k(\chi)$, while the radial geodesic distance remains χ. Then $d_A = D_M/(1+z)$ and $d_L = (1+z)D_M$. Using D_C instead of D_M in those two formulas is a common but avoidable mistake.

$$\boxed{d_L = (1+z)^2 d_A} \tag{33}$$

is **Etherington's distance-duality relation**. It holds when photons travel on null geodesics, their number is conserved, and the spacetime description is metric; it does not require spatial flatness.

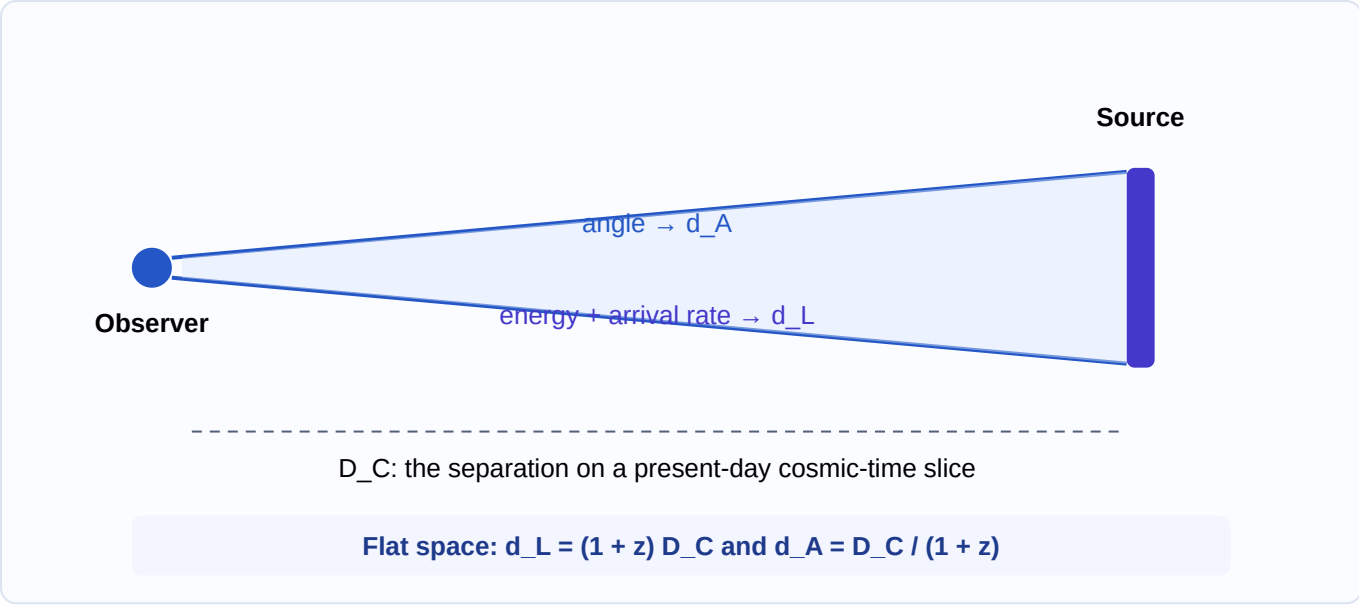


Figure 4. The same source can have different distances because a flux measurement and an angular-size measurement ask different questions.

Why luminosity distance has two factors of $1+z$

Suppose a source emits bolometric luminosity L. Today its photons are spread over area $4\pi D_M^2$. Each photon has lost an energy factor $1/(1+z)$, and the arrival rate is lower by another factor $1/(1+z)$. Thus

$$F = \frac{L}{4\pi D_M^2 (1+z)^2} \equiv \frac{L}{4\pi d_L^2}. \tag{34}$$

This is a **bolometric** statement. A real detector measures a finite band. Comparing emitted and observed bands requires a filter-dependent **K-correction**, and sometimes a luminosity-model correction. A redshifted spectrum does not keep the same flux inside a fixed blue filter.

Why angular size need not shrink forever

For a ruler of proper transverse length ℓ at emission, the small-angle relation is $\theta = \ell/d_A$. As redshift increases, D_M grows but the universe was smaller at emission. The competition can make d_A reach a maximum and then decrease. Beyond that redshift a ruler of fixed proper size can subtend a **larger** angle. There is no contradiction: the light left when the ruler was much closer in proper separation.

Surface brightness provides another powerful check. Bolometric flux falls as d_L^{-2} while apparent solid angle is set by d_A^{-2}. Therefore

$$I_{\rm obs} = \frac{I_{\rm em}}{(1+z)^4}, \tag{35}$$

for corresponding bolometric surface brightness, before practical bandpass effects. This **Tolman dimming** is a consequence of expansion and distance duality, not simply dust making galaxies fainter.

The distance ladder and standard rulers

Parallax calibrates distances geometrically. Those distances calibrate stellar indicators such as Cepheids, which can calibrate Type Ia supernovae in nearby host galaxies. Supernovae then extend the relative distance scale to much larger redshifts. Different rungs overlap; no single method spans the entire range. Other routes, including gravitational-lens time delays and the early-universe sound horizon, test the calibration independently.

Type Ia supernovae are **standardizable**, not identical candles. Light-curve shape and colour help reduce the scatter in inferred luminosity. Dust, population evolution and calibration systematics still matter. Baryon acoustic oscillations (BAO) provide a comoving standard ruler; a measured angle constrains $r_d/D_M(z)$, while a redshift separation constrains roughly $H(z)r_d/c$. The absolute ruler scale r_d depends on early-universe physics.

For a calibrated luminosity distance,

$$m - M = 5\log_{10}\left(\frac{d_L}{10\ \mathrm{pc}}\right) = 5\log_{10}\left(\frac{d_L}{\mathrm{Mpc}}\right) + 25. \tag{36}$$

A constant fractional distance error is a constant magnitude offset. At low redshift, where $d \simeq cz/H_0$, an unknown candle calibration trades directly against H_0. The **shape** of a supernova Hubble diagram can reveal acceleration even when the absolute normalization is uncertain.

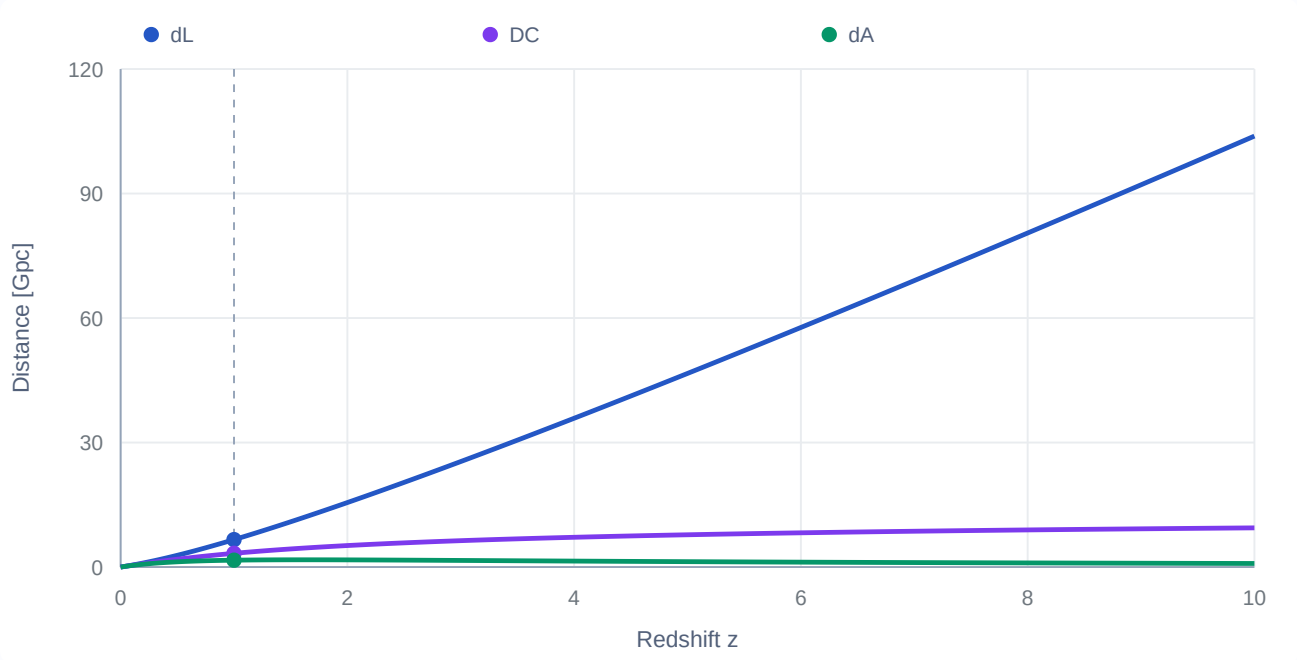
INTERACTIVE EXPERIMENT 4

Ask which distance you mean.

Flat benchmark: $H_0 = 70$, $\Omega_{m0} = 0.3$, $\Omega_{\Lambda 0} = 0.7$; radiation is neglected. The curves are computed by numerical integration, not by cz/H_0. Horizon values are present-day proper distances for this same model.

Comoving / present proper	Luminosity distance	Angular-diameter distance
3.304 Gpc	**6.608 Gpc**	**1.652 Gpc**

Lookback time	Present recession rate HD_0 / c	Particle / event horizon
7.715 Gyr	**0.771**	**46.2 / 15.9 Gly**



At z = 1.00, light has travelled for 7.72 Gyr. The present proper separation is 3.30 Gpc, not c times the lookback time. The present recession rate is 0.77 c.

Try this. Compare z = 1 and z = 3. The luminosity distance keeps increasing, but the angular-diameter distance turns over. Find the approximate turnover by moving the slider. Why does a more distant ruler not always look smaller?

This is a snapshot of the interactive experiment. Open the HTML edition to change its controls.

8. The thermal universe and the CMB

Why expansion cools radiation

A blackbody spectrum is specified by a temperature. For freely streaming photons, every frequency scales as $\nu \propto a^{-1}$. The Planck distribution keeps its shape if the temperature also scales as a^{-1}:

$$B_\nu(T) = \frac{2h\nu^3}{c^2}\frac{1}{e^{h\nu/k_{\rm B}T}-1}, \qquad \boxed{T(z) = T_0(1+z)}. \tag{37}$$

You can see the cancellation directly: replace ν by $\nu/(1+z)$ and T by $T/(1+z)$. The exponential is unchanged, while the prefactor supplies the intensity dilution. The occupation of photon states is preserved; a thermal spectrum does not become an arbitrary collection of lower-energy photons merely because the universe expands.

The total radiation energy density and photon number density obey

$$u_\gamma = a_{\rm rad}T^4 = \frac{4\sigma_{\rm SB}}{c}T^4, \qquad n_\gamma = \frac{2\zeta(3)}{\pi^2}\left(\frac{k_{\rm B}T}{\hbar c}\right)^3 \simeq 20.29\,T^3\ \mathrm{cm^{-3}}. \tag{38}$$

At $T_0 = 2.725\ \mathrm{K}$, $u_\gamma \simeq 4.17\times10^{-14}\ \mathrm{J\,m^{-3}}$, $n_\gamma \simeq 410\ \mathrm{cm^{-3}}$, and the mean photon energy is $u_\gamma/n_\gamma \simeq 2.701 k_{\rm B}T$. The peak of B_λ is at $b_{\rm W}/T \simeq 1.064\ \mathrm{mm}$. The peak of B_ν occurs at a different wavelength when plotted against wavelength: a peak is a property of the chosen spectral variable, not a unique photon colour.

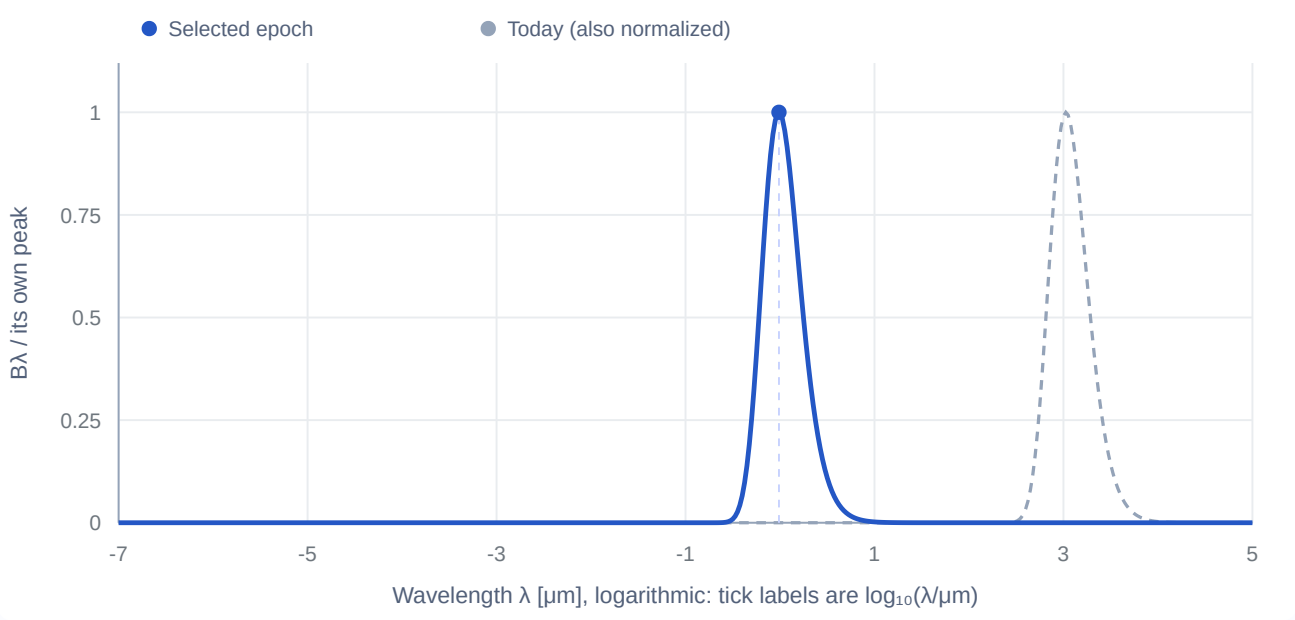
INTERACTIVE EXPERIMENT 3

Watch a blackbody cool without losing its shape.

The solid blue curve is the selected epoch; the dashed grey curve is today. Both are normalized to their own peak so that the shift is visible across many orders of magnitude. Their actual intensities are very different.

Redshift z	Temperature	Bλ peak wavelength
1.09 × 10^3	**2972.8 K**	**0.975 μm**

Energy density / today
1.42 × 10^12



1 + z = 1.09 × 10^3. T = 2972.8 K. Both plotted curves have unit peak height by design; the actual energy density is 1.42 × 10^12 times today's.

Try this. Choose last scattering. Was the CMB then microwave-coloured? Move to the BBN temperature scale and compare the wavelength shift with the energy-density ratio. The BBN preset is a temperature scale, not a claim that a free-streaming CMB already existed then.

This is a snapshot of the interactive experiment. Open the HTML edition to change its controls.

Recombination is not the first moment photons existed

At high temperature, matter is ionized and photons scatter frequently from free electrons. As the universe cools, electrons and nuclei combine into neutral atoms. This process is called **recombination**, although the electrons and nuclei had not previously been combined. Around $z \simeq 1090$, the scattering rate falls enough for most CMB photons to stream freely. The **last-scattering surface** is a broad statistical shell in our past, not a material wall.

A crude Saha equilibrium estimate for hydrogen is

$$\frac{x_e^2}{1-x_e} = \frac{1}{n_{\mathrm{H}}}\left(\frac{m_e k_{\mathrm{B}} T}{2\pi\hbar^2}\right)^{3/2} e^{-13.6\ \mathrm{eV}/k_{\mathrm{B}}T}, \tag{39}$$

where x_e is the ionized fraction under the simplified assumptions. It illustrates why recombination occurs near a few thousand kelvin rather than at $13.6\ \mathrm{eV}/k_{\mathrm{B}} \simeq 1.58 \times 10^5\ \mathrm{K}$: there are many photons per baryon, and the high-energy tail can keep hydrogen ionized. A precision calculation also follows the rates of reactions and the loss of equilibrium; the detailed ionization history is not obtained by putting $k_{\mathrm{B}}T = 13.6\ \mathrm{eV}$.

At last scattering, $T \simeq 2.725 \times 1091 \simeq 2970\ \mathrm{K}$. The photons were then near optical/infrared wavelengths. Expansion has moved that spectrum into microwaves. The CMB is therefore a relic of a hot, opaque phase—not light emitted by a cold shell at today's 2.725 K.

A thermal bath with a few fingerprints

The CMB is astonishingly close to a blackbody and nearly isotropic. Its largest temperature pattern is a **dipole** associated mainly with our motion relative to the CMB rest frame. For small speed, $\Delta T/T \simeq (v/c)\cos\theta$. After subtracting the dipole and foreground emission, intrinsic fluctuations are of order 10^{-5}. Those tiny variations are the seeds and tracers of cosmic structure.

Do not confuse three statements:

- **Spectral:** the distribution of intensity with frequency is nearly Planckian.
- **Isotropic:** the temperature is nearly the same in different directions.
- **Homogeneous:** the underlying large-scale matter distribution is statistically uniform.

They are related through cosmological physics, but they are not interchangeable measurements.

The light-element recipe

During the first several minutes, nuclear reactions produced much of the universe's helium-4, some deuterium and helium-3, and small amounts of lithium-7. This is **Big Bang nucleosynthesis (BBN)**. Most hydrogen nuclei survived as hydrogen. The heavier elements were made mainly later, in stars and stellar explosions or mergers.

A useful sequence is:

1. At high temperature, weak interactions interconvert neutrons and protons.
2. As the temperature falls, the neutron-to-proton ratio decreases and weak reactions freeze out.
3. Free neutrons decay while stable deuterium formation is delayed by energetic photons—the **deuterium bottleneck**.
4. Once deuterium survives, reactions rapidly assemble helium-4.
5. Expansion and falling density stop efficient fusion before large amounts of heavier elements form.

If the neutron-to-proton ratio just before nucleosynthesis is $r = n_n/n_p$, and essentially every available neutron enters helium-4, the helium mass fraction is

$$Y_p = \frac{4n_{\mathrm{He}}}{n_{\mathrm{H}} + 4n_{\mathrm{He}}} = \frac{2r}{1+r}. \tag{40}$$

For $r = 1/7$, $Y_p = 1/4$. This is a counting result, not a complete nuclear-reaction network. The baryon-to-photon ratio $\eta = n_b/n_\gamma \sim 6 \times 10^{-10}$ controls how efficiently reactions occur. Deuterium is particularly sensitive to the baryon density; the observed abundances provide a remarkable consistency test with the CMB. The predicted and observed lithium-7 abundances do not agree perfectly—the unresolved **lithium problem** is a reason to distinguish a successful framework from a claim that every detail is solved.

Thermal milestones, approximately

Epoch	Redshift / temperature scale	Physical change
early radiation era	model-dependent	relativistic particles dominate the energy density
BBN	$T \sim 10^9$ K, first minutes	light nuclei form
matter–radiation equality	$z \sim 3400$ in the real universe	matter and radiation densities become equal
recombination / last scattering	$z \simeq 1090$, age about 0.38 Myr	CMB photons begin free streaming
first stars and reionization	broadly $z \sim 6$–20, source-dependent	ultraviolet light reionizes intergalactic gas
late-time acceleration	benchmark $z \simeq 0.67$	$\ddot{a}$ changes sign

The order of BBN and recombination matters: **nuclei form first; neutral atoms form much later.** The numerical ages depend on the cosmological model, and “first stars” is not a single sharply defined instant.

9. Reading the CMB and the matter distribution

From a map to a power spectrum

Write the fractional temperature fluctuation in direction $\hat{\mathbf{n}}$ as $\Delta T(\hat{\mathbf{n}})/T$. Decompose it into spherical harmonics:

$$\frac{\Delta T}{T}(\hat{\mathbf{n}}) = \sum_{\ell m} a_{\ell m} Y_{\ell m}(\hat{\mathbf{n}}), \qquad C_\ell = \left\langle |a_{\ell m}|^2 \right\rangle. \tag{41}$$

You do not need to memorize the special functions to understand the question: **how much fluctuation is present on each angular scale?** Large ℓ means small angular separation, roughly $\theta \sim 180^\circ/\ell$. A plotted quantity is often $\ell(\ell+1)C_\ell/(2\pi)$ rather than C_ℓ itself. Always read the axes before comparing peak heights.

Before photons decouple, gravity pulls the photon–baryon fluid into overdensities while photon pressure resists compression. The result is acoustic oscillations. Modes caught at different phases produce a series of

peaks. The sound horizon at the drag epoch is the characteristic comoving distance a pressure wave could travel:

$$r_s = \int_0^{t_{\rm drag}} \frac{c_s(t)}{a(t)} dt, \qquad c_s \simeq \frac{c}{\sqrt{3(1+R)}}, \qquad R = \frac{3\rho_b}{4\rho_\gamma}. \tag{42}$$

The precise BAO ruler r_d and the CMB sound horizon r_* refer to closely related but not identical epochs. They are both of order $150\ \mathrm{Mpc}$ in the standard model; do not replace one by the other in a precision calculation without checking the definition.

What different features tell us

- **Angular acoustic scale:** the ratio of a known early-universe ruler to its distance tests geometry and expansion history. The first peak near $\ell \sim 200$ is consistent with nearly flat spatial geometry in the full model.
- **Relative peak heights:** baryons add inertia to the photon–baryon fluid, changing compression and rarefaction peaks. Matter density affects the gravitational potentials and the epoch of equality.
- **Damping tail:** photon diffusion smooths very small-scale fluctuations, producing Silk damping. The finite thickness of last scattering also matters.
- **Large-angle patterns:** primordial fluctuations, gravitational potential effects and the integrated evolution of potentials all contribute. There is no single “Doppler-only” explanation of the entire map.
- **Polarization:** Thomson scattering with a quadrupolar incident radiation field generates linear polarization. Its pattern can be separated into E and B components. Lensing converts some E into B; primordial gravitational waves could also produce B modes, but a B-mode detection would require careful treatment of dust and other foregrounds.

A single beautiful CMB image is not a measurement of every cosmological parameter by itself. Parameter constraints come from a model fitted to spectra, with foreground cleaning, instrumental calibration, likelihoods and degeneracies.

Baryon acoustic oscillations as a ruler

The same early pressure waves leave a preferred separation in the late-time matter distribution. It is not a sharp shell around every galaxy. It is a small statistical excess in pair separations—a feature in the correlation function or a sequence of wiggles in the power spectrum.

For a transverse BAO measurement,

$$\theta_{\rm BAO} \simeq \frac{r_d}{D_M(z)}. \tag{43}$$

For a radial measurement, a small redshift interval obeys

$$\Delta z \simeq \frac{H(z) r_d}{c}. \tag{44}$$

Thus BAO can constrain distances and $H(z)$, but the absolute scale requires a calibrated r_d. This is why BAO and supernova data are complementary: one is particularly sensitive to a physical ruler, the other to relative luminosity distances.

10. Dark matter, dark energy and the growth of structure

The words sound similar because both name inferred ingredients rather than directly identified laboratory substances. Their roles are very different.

Question	Cold dark matter	Cosmological constant / dark energy
Clustering	clusters gravitationally	approximately smooth in ΛCDM
Pressure	approximately zero	$P = -u$ for Λ
Density scaling	a^{-3}	constant for Λ
Main role here	helps structure grow	drives late-time acceleration
Typical evidence	rotation curves, lensing, clusters, CMB and structure	supernova distances combined with CMB/BAO and other probes

Dark matter is not simply “dark ordinary gas.” BBN, the CMB and structure formation constrain the total baryon density and favour a dominant nonbaryonic matter component. Its particle identity remains unknown. Modified-gravity alternatives must address the full set of observations, not just one galaxy rotation curve.

A simple growth calculation

Define the matter density contrast

$$\delta(\mathbf{x}, t) = \frac{\rho_m(\mathbf{x}, t) - \bar{\rho}_m(t)}{\bar{\rho}_m(t)}. \tag{45}$$

On subhorizon scales, after radiation pressure effects are negligible, linearized Newtonian gravity gives

$$\ddot{\delta} + 2H\dot{\delta} - 4\pi G \bar{\rho}_m \delta = 0. \tag{46}$$

The three terms say: the overdensity accelerates inward, expansion supplies a drag-like term, and gravity amplifies the contrast. In an Einstein–de Sitter universe, try $\delta \propto t^n$. Since $H = 2/(3t)$ and $4\pi G\bar{\rho}_m = 2/(3t^2)$,

$$n(n-1)+\frac{4}{3}n-\frac{2}{3}=0 \quad \Rightarrow \quad n=\frac{2}{3},\ -1. \tag{47}$$

The growing mode is $\delta \propto t^{2/3} \propto a$. Linear growth is slow, not an instantaneous collapse. Once $\delta \sim 1$, nonlinear collapse and virialization require more than this equation. During strong radiation domination or Λ domination, matter growth is less efficient. Dark energy therefore affects not only distances but also the history of structure.

The observed cosmic web is a major constraint. A universe with only the tiny baryonic perturbations expected without an additional gravitating matter component has difficulty reaching today's structure by the available time. Modern analyses quantify the clustering amplitude with parameters such as σ_8 and $S_8=\sigma_8\sqrt{\Omega_{m0}/0.3}$; these are model- and scale-dependent summaries, not direct photographs of “the amount of dark matter.”

11. Inflation and the limits of the hot Big Bang

Two puzzles in ordinary decelerating expansion

Horizon puzzle. Widely separated regions of the last-scattering surface have nearly the same temperature. In a universe that was always radiation- or matter-dominated after its beginning, many of those regions would not have had time to exchange a signal. Why do they share such similar conditions?

Flatness puzzle. The curvature contribution relative to the total density obeys

$$|\Omega_{\rm tot}-1|=\frac{|k|c^2}{a^2H^2}. \tag{48}$$

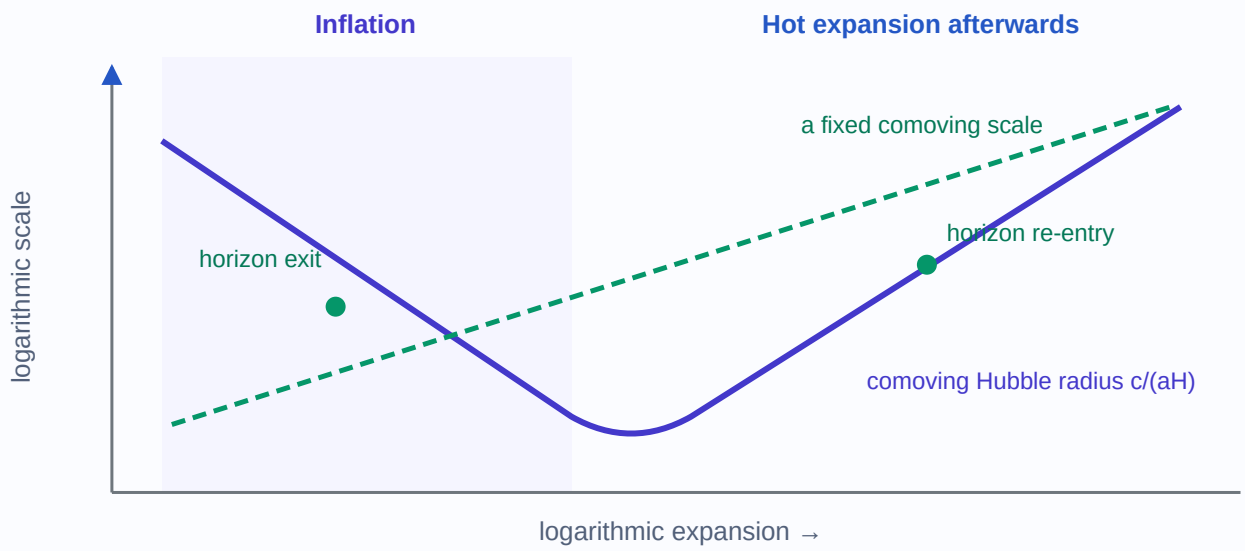
During radiation domination, $aH \propto a^{-1}$ decreases, so the fractional curvature deviation grows approximately as a^2. During matter domination it grows approximately as a. A universe close to flat today would have had to be extraordinarily close to flat in the early decelerating era. This is a question about initial conditions, not a logical contradiction.

The key idea: shrink the comoving Hubble radius

Inflation is a proposed early interval of accelerated expansion,

$$\ddot{a}>0 \quad \Longleftrightarrow \quad \frac{d}{dt}\left(\frac{c}{aH}\right)<0. \tag{49}$$

Physical wavelengths grow with a, while the Hubble radius c/H changes comparatively slowly in near-exponential expansion. A small region can be stretched until it encompasses the scales that later become our observable universe. If that region was causally connected before inflation, the horizon puzzle is alleviated. At the same time, increasing aH suppresses $|\Omega_{\rm tot}-1|$.



Schematic trends; slopes and transition sharpness are not a fitted model.

Figure 5. Inflation stretches a small causal patch. The same shrinking comoving Hubble radius suppresses the observable curvature deviation.

The number of **e-folds** is

$$N = \ln \frac{a_{\rm end}}{a_{\rm start}} = \int H\, dt. \tag{50}$$

For constant H, $N = H\Delta t$. Sixty e-folds means a linear stretch by $e^{60} \sim 10^{26}$, not by 60. The required number is model-dependent because reheating and the inflationary energy scale affect the mapping from today's scales back to inflation. “Exactly 60” is a useful estimate, not a universal constant.

Extension — a scalar-field realization

A homogeneous scalar field with canonical kinetic energy has

$$u_\phi = \frac{1}{2}\dot\phi^2 + V(\phi), \qquad P_\phi = \frac{1}{2}\dot\phi^2 - V(\phi), \tag{51}$$

in units with $c = 1$ and a suitably normalized field. If the potential dominates, $P_\phi \simeq -u_\phi$ and the field behaves approximately like a cosmological constant. Its equation of motion is

$$\ddot\phi + 3H\dot\phi + V_{,\phi} = 0. \tag{52}$$

The $3H\dot\phi$ term acts like friction. Under slow-roll conditions, define

$$\epsilon_H = -\frac{\dot H}{H^2} < 1, \qquad \epsilon_V \simeq \frac{M_{\rm Pl}^2}{2}\left(\frac{V_{,\phi}}{V}\right)^2, \tag{53}$$

where the second expression uses the reduced Planck mass and the slow-roll approximation. Inflation ends when the accelerated-expansion condition fails. **Reheating** transfers energy from the inflationary sector into particles and radiation, connecting this episode to the hot Big Bang. The details are not known.

Quantum fluctuations of the field and metric can be stretched into classical-looking primordial perturbations. Many models predict nearly scale-invariant, nearly Gaussian, predominantly adiabatic fluctuations. A small departure from exact scale invariance is described by a spectral index n_s, with $n_s = 1$ meaning exact scale invariance for the conventional scalar spectrum. Tensor fluctuations are possible, but their amplitude is model-dependent.

> ***What inflation does not say.*** It is not an explosion into empty space. It does not mean every part of the entire universe must be exactly uniform. It does not prove what happened at a mathematical first instant. Nor is every inflationary model confirmed: the framework addresses important initial-condition problems, while particular potentials, reheating histories and primordial gravitational-wave signals remain under test.

12. Horizons: what can be seen, and what can be reached?

Three lengths are often confused because each is measured in billions of light-years.

1. **Hubble radius:** $R_H = c/H$, an instantaneous expansion scale.
2. **Particle horizon:** the greatest present proper distance from which light emitted in the earliest accessible past could have reached us.
3. **Cosmic event horizon:** the greatest present proper distance from which a signal emitted now could ever reach us in the future.

For a flat universe, the present particle-horizon distance and event-horizon distance are

$$D_{\mathrm{ph},0} = c\int_0^{t_0}\frac{dt}{a(t)} = \frac{c}{H_0}\int_0^{\infty}\frac{dz}{E(z)}, \tag{54}$$

$$D_{\mathrm{eh},0} = c\int_{t_0}^{\infty}\frac{dt}{a(t)} = \frac{c}{H_0}\int_{-1}^{0}\frac{dz}{E(z)}, \tag{55}$$

provided the integrals converge. In the second expression, $E(z)$ means the analytic continuation of the specified expansion model into the future. If the integral diverges, there is no finite event horizon of that type.

For the rounded matter-plus-Λ benchmark, $D_{\mathrm{ph},0}$ is roughly $14\ \mathrm{Gpc}$, or $46\ \mathrm{Gly}$, while $D_{\mathrm{eh},0}$ is roughly $4.9\ \mathrm{Gpc}$, or $16\ \mathrm{Gly}$. These values are model-dependent. The particle horizon is not ct_0: the distances between the emitting matter and us changed while the light was travelling.

Three clean examples

- **Einstein–de Sitter:** $D_{\rm ph} = 3ct = 2c/H$. There is no finite future event horizon; given enough time, light can cross any fixed comoving separation.
- **Pure de Sitter with constant H:** a signal emitted now can reach us only if its present proper distance is below c/H. The Hubble radius and event horizon coincide in this special model.
- **Our approximate matter-plus-Λ model:** the Hubble radius, particle horizon and event horizon are all different. As Λ comes to dominate, the event horizon approaches an approximately constant proper size while the comoving horizon shrinks.

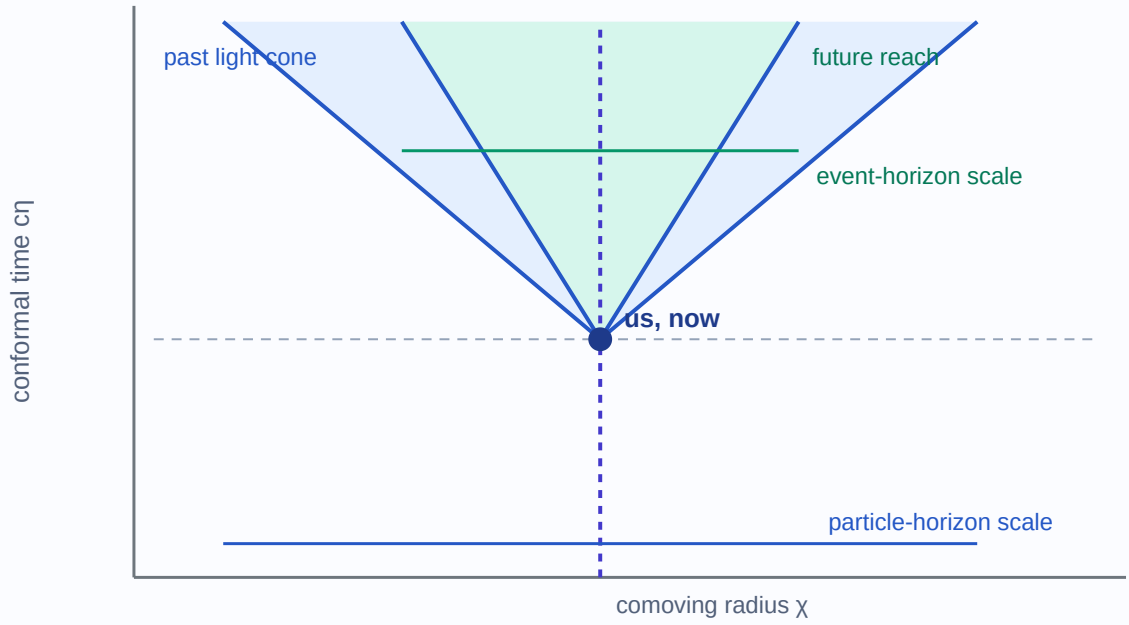


Figure 6. Radial light rays are straight in conformal coordinates. The particle horizon looks backward; the event horizon looks forward.

Can we see galaxies receding faster than light?

Yes. The rate HD is the change of a proper distance defined across a cosmic-time slice. It is not a locally measured speed past a nearby observer, so special relativity's local speed limit is not violated. Light always moves locally at c. Whether a photon reaches us depends on the entire future behaviour of the intervening expansion.

For an incoming radial light ray, its proper distance D from us obeys

$$\dot{D} = HD - c. \tag{56}$$

If it begins outside the Hubble radius, D can initially increase even though the photon is aimed toward us. If the comoving Hubble radius later grows enough, the photon can enter the region where $HD < c$ and then approach us. This is how a “Hubble crossing” can occur. The Hubble sphere is not a one-way visibility wall.

Extension — conformal time makes light rays straight

Define conformal time by $d\eta = dt/a(t)$. The radial FLRW metric becomes

$$ds^2 = a^2(\eta)\left[-c^2 d\eta^2 + d\chi^2 + S_k^2(\chi) d\Omega^2\right]. \tag{57}$$

Radial light obeys $d\chi = \pm c\, d\eta$, so it travels on straight 45-degree lines when the axes are χ and $c\eta$. The particle horizon is the amount of conformal time accumulated since the beginning; the event horizon is the amount remaining. This turns a difficult-sounding causal question into an area-under-a-curve question.

13. Measuring cosmology without fooling yourself

The low-redshift Hubble diagram

At sufficiently low redshift, $cz \simeq H_0 D$ after correcting to an appropriate cosmic reference frame. A galaxy's peculiar velocity adds scatter. For a line constrained through the origin, the least-squares slope is

$$\widehat{H}_0 = \frac{\sum_i D_i v_i}{\sum_i D_i^2}, \qquad v_i \simeq c z_i. \tag{58}$$

If the velocity uncertainties are unequal, use inverse-variance weights. Averaging the individual ratios v_i/D_i is not the same estimator and can give very noisy nearby galaxies excessive influence. A real analysis also treats distance uncertainties, correlated peculiar motions, selection effects, reference-frame corrections and calibration errors.

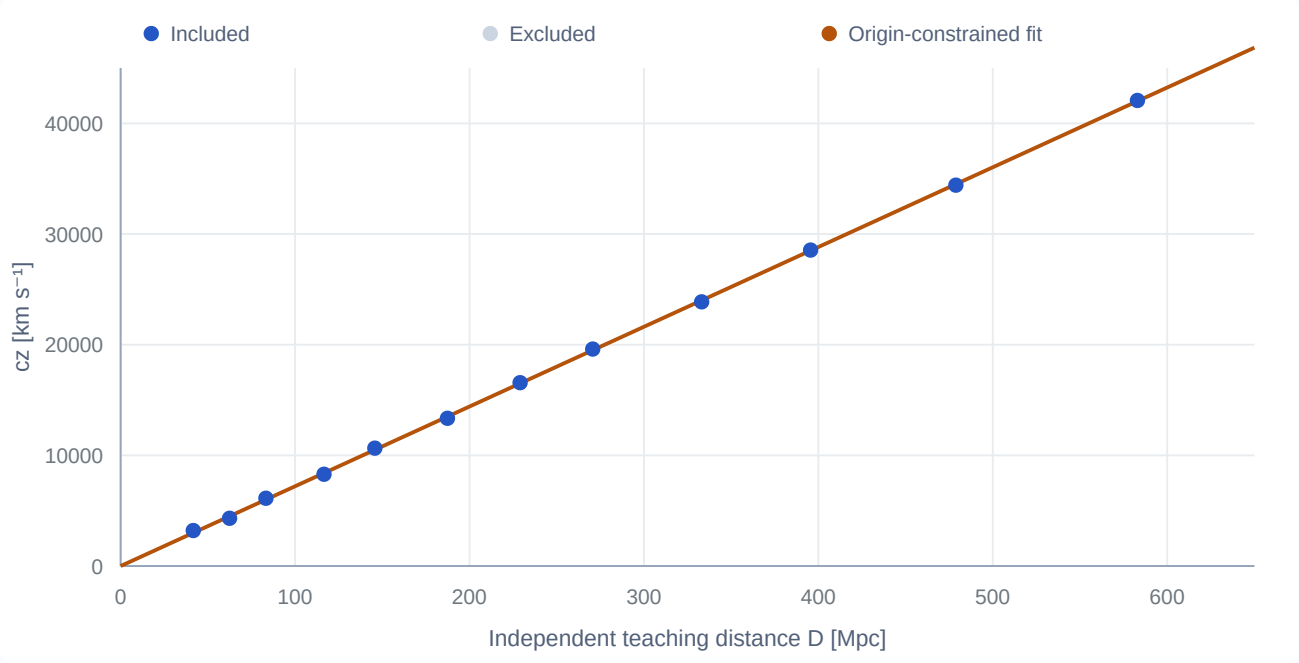
INTERACTIVE EXPERIMENT 5

Fit the Hubble law—and notice what the fit cannot tell you.

These twelve observations are **synthetic teaching data**, generated with $H_0 = 72$ and fixed velocity perturbations. The fit is constrained through the origin with equal velocity uncertainties. It is not a measurement of the real universe.

Fitted H_0 [km s^{-1} Mpc^{-1}]	Formal statistical error	Galaxies included
72.06	**±0.20**	**12**

Velocity RMS residual
127.8 km s^{-1}



Synthetic-data fit: $H_0 = 72.06 \pm 0.20$ km s^{-1} Mpc^{-1} (formal statistical error only), using 12 galaxies. The generating value was 72; this is not an observational result.

Try this. Exclude the nearest galaxies. Does the scatter decrease? The formal error does not include a shared distance-calibration error or correlated bulk flow. Why should those be reported separately?

This is a snapshot of the interactive experiment. Open the HTML edition to change its controls.

Expansion, acceleration and supernova magnitudes

For a smooth distance–redshift relation in a flat universe,

$$D_C(z) = \frac{c}{H_0}\left[z - \frac{1+q_0}{2}z^2 + O(z^3)\right], \tag{59}$$

$$d_L(z) = \frac{c}{H_0}\left[z + \frac{1 - q_0}{2}z^2 + O(z^3)\right]. \tag{60}$$

The second-order term contains the acceleration information. A single distant supernova cannot establish acceleration: one needs a calibrated population, a redshift baseline and a model for systematics. Relative supernova distances can constrain the shape of the expansion history even when the absolute value of H_0 is marginalized over.

A fractional distance error ϵ gives a magnitude shift

$$\Delta\mu = 5\log_{10}(1+\epsilon) \simeq \frac{5}{\ln 10}\epsilon. \tag{61}$$

Thus a 1% distance-calibration error is about 0.0217 mag. Small magnitude offsets matter. If all supernova absolute magnitudes are shifted by ΔM, the inferred low-redshift Hubble constant changes according to

$$\frac{H_{0,\mathrm{fit}}}{H_{0,\mathrm{true}}} = 10^{\Delta M/5}, \tag{62}$$

with the sign defined as assumed minus true absolute magnitude. This is a calibration degeneracy, not evidence that the universe changes when we relabel a candle.

What a “tension” means

Different measurements of H_0 can disagree more strongly than their quoted uncertainties would suggest. This is called the **Hubble tension**. It could indicate unrecognized systematics, a missing ingredient in the assumed model, or a combination of effects. It is not permission to declare any preferred value correct.

A useful comparison asks:

1. Are the measurements genuinely independent, or do they share a calibration?
2. Is the inferred quantity model-dependent?
3. Are uncertainties statistical, systematic, or correlated?
4. Does a proposed new model improve the full data set, or only the one discrepancy?
5. How many extra parameters were introduced to obtain the improvement?

The minimal ΛCDM model is remarkably economical and successful, but it is not a final theory. It does not identify the dark matter particle, explain the microscopic origin of Λ, or supply a tested quantum theory of the earliest epoch. Observations can also test whether the accelerating component evolves with time. A constant $w = -1$ is a benchmark hypothesis, not a logical necessity. This handout uses it because its consequences can be calculated cleanly.

A compact chain of inference

Observation	Directly measured	Inference requiring a framework
galaxy spectra	line wavelengths and shifts	expansion history after distances and peculiar motions are treated
supernovae	fluxes, colours, light curves	relative distances and acceleration
CMB	intensity, frequency spectrum, anisotropies	contents, geometry and primordial initial conditions
BBN spectra/abundances	light-element abundances	baryon density and early expansion conditions
galaxy clustering	positions, redshifts and shapes	growth, bias and the matter power spectrum
gravitational lenses	image positions, shapes and time delays	mass distribution and distance combinations

14. The Olympiad problem-solving toolkit

First decide what kind of quantity you have

- A **redshift** gives a ratio of scale factors.
- A **lookback time** requires an integral over expansion history.
- A **flux** calls for luminosity distance and bandpass care.
- An **angular size** calls for angular-diameter distance.
- A **causal limit** calls for a horizon integral, not just c/H.
- A **density comparison** needs the appropriate power of $1+z$.

Write the definition before substituting numbers. Most difficult cosmology problems become manageable when the correct distance or epoch is identified.

A reliable five-step method

1. **Draw the events:** emission, observation, equality or acceleration. Mark a, z and t separately.
2. **Choose the model:** matter-only, radiation-only, flat matter-plus-Λ, constant H, or a supplied $H(z)$. State neglected components.
3. **Find the governing relation:** conservation, Friedmann, distance duality, a null path or a horizon integral.
4. **Check limits and dimensions:** $z \to 0$, $a \to 1$, $\Omega_\Lambda \to 0$, equal densities, or a pure-component universe.

5. **Interpret the result:** is it a time, a present proper distance, an emission-time distance, or an observable angle? Explain what the number means.

Essential results, with their conditions

Result	Conditions / warning
$D = a\chi$	flat geometry, comoving separation
$1 + z = a_0/a_{\mathrm{em}}$	smooth-background cosmological redshift
$E^2 = \sum_i \Omega_{i0}(1+z)^{3(1+w_i)} + \Omega_{k0}(1+z)^2$	FLRW; separate non-interacting constant-w fluids
$u \propto a^{-3(1+w)}$	local conservation, constant w, no energy exchange
$d_L = (1+z)^2 d_A$	photon conservation and metric propagation; not restricted to flat space
$t_0 = 2/(3H_0)$	flat, matter-only throughout
$T = T_0(1+z)$	freely cooled radiation / thermal CMB temperature
$D_{\mathrm{ph}} = 3ct$	Einstein–de Sitter, not ΛCDM at all times
$D_{\mathrm{eh}} = c/H$	pure de Sitter with constant H, not every cosmology

Common mistakes that cost marks

- Treating cz as a special-relativistic velocity at every redshift.
- Using today's CMB temperature in a last-scattering calculation.
- Forgetting the arrival-rate factor in luminosity distance.
- Comparing an emission-time ruler with a present-day angular-diameter distance.
- Saying matter–Λ equality is the start of acceleration.
- Assuming positive curvature forces recollapse even with Λ.
- Treating the Hubble radius as an absolute observational boundary.
- Calling the Big Bang an explosion from a point in space.
- Mixing mass density ρ with energy density u.
- Applying a matter-only age formula to a radiation-era problem.
- Adding a cosmological-constant term twice, once as Λ and once as a fluid.
- Quoting five significant figures from two-significant-figure input data.

A final physical picture. Expansion changes the relation between coordinates and rulers. Light carries a record of that change. Gravity and pressure determine how the expansion rate evolves. The contents cool, decouple and cluster. Our observations are a past-light-cone view of that history, not a snapshot of the entire universe today. Keep those five sentences in mind and the formulas become a connected story rather than a list to memorize.

15. Sources, attribution and further exploration

The explanations, diagrams, simulations and original exercises in this handout were prepared for MAIS PAC. The problem set distinguishes **website-sourced adaptations**, **original IOAA-inspired challenges**, and **original club foundations**. Adaptations are not official IOAA questions, and the worked solutions are independent explanations, not official marking schemes.

Website problem sources beyond the IOAA archive

- University of Sheffield, PHY306: Introduction to Cosmology — Practice Problems: distance definitions, systematic errors, fluid scaling, CMB and inflation. Problems 4 and 3 are adapted here; Problems 6, 13 and 43 provide further practice on the same concepts.
- Yale University, ASTR 610: Problem Set 1: Einstein–de Sitter cosmology, radiation equality, tired light and curved geometry. Problems 1–4 provide source material for several exercises.
- MIT OpenCourseWare, 8.286 Fall 2013: Quiz 3 Review Problems: Problem 9 on flatness and Problem 10 on distance definitions at high redshift informed the extensions. The source's historical “most distant quasar” wording is not presented as a current record.
- BAAO 2015 Round 2, Question 12 and official solutions: a school-to-international-level redshift and expansion problem. Our version supplies its own assumptions and independently worked answers.

IOAA connections, with direct materials

- IOAA 2022, Theory 4, “Journey Between Galaxies”: motion relative to an expanding background.
- IOAA 2023, Theory 5, “Dark Energy”: critical density, dark-energy density and matter–Λ equality.
- IOAA 2024, Theory 5, “CMB”: Wien scaling and an explicitly matter-dominated age estimate.
- IOAA 2020, Team 14, “Spectral Distortions of the CMB”: blackbody radiation and energy injection. Our simplified fixed-photon-number exercise is not the full thermalization problem.
- IOAA 2025, Theory 10, “Big Bang Nucleosynthesis”: the topic inspiration for the neutron-counting extension; the exercise here is independently constructed.
- Official IOAA past-materials archive and searchable year-by-year collection.

For deeper study, use Barbara Ryden's *Introduction to Cosmology*, Steven Weinberg's *Cosmology*, and the public lecture notes of MIT 8.286. The first develops the physical picture; the later sources go further into relativity and perturbations. The purpose of the extension sections is to make those texts approachable—not to pretend that a short handout replaces them.

16. Lecture + Problems: how to use the set

There are **30 problems**: twelve foundations, eight explicitly sourced website adaptations, and ten original Olympiad-level challenges. Difficulty follows the handout convention: **[1]** is a warm-up, **[2]** requires a multi-step Olympiad argument, and **[3]** is a challenge or mathematical extension. The extensions are enrichment beyond the minimum IOAA requirement to avoid differential equations.

Use the constants in Section 1 unless a problem supplies different values. Keep units until the last line. A calculator is useful; a computer is not required except for optional data exploration. Give reasons alongside numerical answers. A correct number without its assumptions is an incomplete cosmological answer.

Suggested routes. **First reading:** Problems 1, 3, 5–9. **Olympiad training:** 2, 4, 10–16, 20–27. **Beyond-syllabus enrichment:** 17–19 and 28–30. For a roughly 90-minute mock paper, try 3, 7, 8, 13, 20 and 22 before opening the solutions. The interactive Hubble laboratory uses a clearly labelled synthetic catalogue, not real survey measurements.

Attribution key. "Adapted from" identifies the source of the underlying exercise; numbers, scaffolding and wording may have been changed. "MAIS original — IOAA-inspired" means an independently constructed problem motivated by the listed past-material topic, not an official or reproduced IOAA question. The Solutions restate the tasks, derive the results and explain the strategy for the difficult originals.

17. Foundations: from units to cosmic history

[1] Problem 1 — A rate disguised as a velocity. *MAIS original foundation.*

(a) Convert $H_0 = 70\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ to $\mathrm{s^{-1}}$ and find H_0^{-1} in Gyr.

(b) Find the Hubble radius c/H_0 in Gpc and Gly.

(c) A student says, "The universe must be 13.97 billion years old." Explain what extra assumption would make that statement valid, and why it is not generally valid.

(d) If the benchmark density parameters are $\Omega_{m0} = 0.3$ and $\Omega_{\Lambda 0} = 0.7$, find the present critical mass density and the mean matter mass density in SI units.

Worked solution →

[2] Problem 2 — When does a galaxy trace the Hubble flow? *MAIS original foundation.*

A galaxy has a line-of-sight peculiar velocity whose magnitude may be as large as $300\ \mathrm{km\,s^{-1}}$. Take $H_0 = 70\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ and use the low-redshift law.

(a) At $D = 2\ \mathrm{Mpc}$, find the Hubble velocity and the largest possible fractional contamination of the inferred velocity.

(b) How far away must a galaxy be for this contamination to be at most 5%?

(c) Explain why a nearby blueshift does not disprove expansion.

(d) Why does averaging many galaxies help with random peculiar velocities but not automatically remove a coherent bulk flow?

Worked solution →

[2] Problem 3 — One redshift, three different questions. *MAIS original foundation.*

A rest-frame line at $486.1\ \mathrm{nm}$ is observed at $1458.3\ \mathrm{nm}$. Ignore peculiar and gravitational shifts.

(a) Find z and the scale factor at emission.

(b) A rest-frame 20-day supernova feature occurs at the same redshift. How long does it appear to last?

(c) If the shift were instead interpreted as a pure special-relativistic radial Doppler effect, what speed would be inferred?

(d) Explain why the answer to (c) is not, by itself, the proper-distance recession rate of a cosmological source.

Worked solution →

[2] Problem 4 — The ruler that did not stretch. *MAIS original foundation.*

Two ideal comoving galaxies have a present proper separation of $12{,}000\ \mathrm{Mpc}$. A galaxy group lies between them and is gravitationally bound.

(a) What was the comoving galaxies' proper separation at $z = 2$?

(b) By what factor did it change between emission and observation?

(c) Does a $1\ \mathrm{m}$ ruler inside the bound group also change by that factor? Explain.

(d) In the flat $(0.3, 0.7)$ benchmark, $H(z = 2) \simeq 2.97 H_0$. Find the Hubble-flow rate corresponding to the galaxies' separation then, in units of c, and state why a value above c would not violate special relativity.

Worked solution →

[1] Problem 5 — Critical does not mean heavy. *MAIS original foundation.*

Use $H_0 = 70\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$.

(a) Derive $\rho_{\mathrm{crit}} = 3H_0^2/(8\pi G)$ from the flat-space Friedmann equation.

(b) Express the corresponding energy density in $\mathrm{J\,m^{-3}}$.

(c) Using $m_p = 1.67 \times 10^{-27}$ kg, express the critical mass density as an equivalent number of proton masses per cubic metre.

(d) Explain why this number is not a count of the actual protons in space.

Worked solution →

[2] Problem 6 — A thermodynamic derivation of three eras. *MAIS original foundation.*

A non-interacting uniform fluid has constant $w = P/u$ in a comoving volume $V \propto a^3$.

(a) Use $d(uV) = -P\,dV$ to derive $u(a)$.

(b) Specialize to matter, radiation and Λ.

(c) At $z = 3$, find the ratios u_m/u_{m0}, u_r/u_{r0} and $u_\Lambda/u_{\Lambda 0}$.

(d) Explain physically why radiation has one extra power of a^{-1}.

(e) For a constant-w component, find the range of w that accelerates a flat universe when that component dominates.

Worked solution →

[2] Problem 7 — The CMB was once warm. *MAIS original foundation.*

Take $T_0 = 2.725$ K, $z_{\mathrm{ls}} = 1090$, and the present photon energy density $u_{\gamma 0} = 4.17 \times 10^{-14}\ \mathrm{J\,m^{-3}}$.

(a) Find the CMB temperature and the wavelength of the peak of B_λ at last scattering.

(b) Find the photon energy density then.

(c) By what factor has the wavelength of an individual freely propagating CMB photon increased?

(d) Why is it wrong to say that the CMB was emitted by a shell that was already at 2.725 K?

(e) Explain why a temperature obtained from the peak of B_ν need not use the same numerical Wien constant.

Worked solution →

[2] Problem 8 — Equality is not acceleration. *MAIS original foundation.*

Use a flat matter-plus-Λ model with $\Omega_{m0} = 0.3$, $\Omega_{\Lambda 0} = 0.7$.

(a) Find the redshift at which the matter and Λ energy densities are equal.

(b) Find the redshift at which $\ddot{a} = 0$.

(c) Which event occurs earlier? Explain the factor of two that distinguishes them.

(d) Find q_0.

(e) Can $H(z)$ decrease while the expansion accelerates? Justify your answer using $q = -1 - \dot{H}/H^2$.

Worked solution →

[2] Problem 9 — A standard candle and a standard ruler. *MAIS original foundation.*

At $z = 1$, a flat model gives $D_C = 3.30\ \mathrm{Gpc}$. A source has bolometric luminosity $L = 4.00 \times 10^{36}\ \mathrm{W}$ and proper transverse diameter $\ell = 20.0\ \mathrm{kpc}$.

(a) Find d_L, d_A and the emission-time proper distance.

(b) Find the observed bolometric flux in SI units.

(c) Find the angular diameter in arcseconds.

(d) Verify distance duality.

(e) State two reasons a real finite-band observation may need corrections beyond this calculation.

Worked solution →

[2] Problem 10 — The age is an integral. *MAIS original foundation.*

Take $H_0 = 70\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$.

(a) Derive the age of a flat matter-only universe.

(b) For a flat matter-plus-Λ model with $(\Omega_{m0}, \Omega_{\Lambda 0}) = (0.3, 0.7)$, write the age as a definite integral and evaluate it using the closed form in Section 6.

(c) Explain physically why the second age is larger even though the present expansion rate is the same.

(d) Why should neither calculation be extrapolated unchanged through the radiation era when asking for a precision age?

Worked solution →

[2] Problem 11 — A circle inside curved space. *MAIS original foundation.*

On a two-dimensional sphere of radius R, a circle is drawn at geodesic radius r from its centre.

(a) Derive its circumference $C(r)$.

(b) Expand the result for $r \ll R$ to the first non-Euclidean correction.

(c) For $R = 6400\ \text{km}$, estimate the radius of a circle whose circumference differs from $2\pi r$ by $1\ \text{m}$.

(d) Explain why the result does not mean that small curved spaces are exactly Euclidean.

Worked solution →

[3] Problem 12 — Three horizons in one model. *MAIS original foundation; extension.*

Consider an Einstein–de Sitter universe.

(a) Derive $D_{\text{ph}}(t) = 3ct$ and express it in terms of c/H.

(b) Does this model possess a finite future event horizon? Prove your answer using a conformal-time integral.

(c) Contrast this with pure de Sitter expansion at constant H, for which the present event horizon is c/H.

(d) Explain why the particle horizon of the benchmark ΛCDM model can exceed both ct_0 and c/H_0.

Worked solution →

18. Website-sourced problems: beyond the IOAA lists

[3] Problem 13 — When do distance definitions matter? *Adapted from Sheffield PHY306, Problem 4.*

(a) Suppose distance measurements have a 5% systematic uncertainty. Using the leading fractional difference between d_L and d_A, estimate the redshift above which confusing these two definitions matters at least as much as that uncertainty.

(b) For a line-of-sight peculiar velocity of $1000\ \text{km}\,\text{s}^{-1}$, find the minimum distance for a velocity contamination below 10% if H_0 lies between 50 and 100 in the usual units. Use the most conservative value.

(c) At that distance, estimate the low-redshift z for both endpoint values of H_0. Is the distinction between d_L and d_A then larger or smaller than 5%?

(d) Explain why the answer to (a) is an estimate rather than a universal sharp boundary.

Worked solution →

[3] Problem 14 — Count the photons, not just their energies. *Adapted from Sheffield PHY306, Problem 3.*

A source emits N photons of emitted-frame mean energy $\bar{E}_{\rm em}$ during proper time $\Delta t_{\rm em}$. Assume photon conservation and a spatially flat background. At observation, the photons are spread over a sphere of area $4\pi D_C^2$.

(a) Derive the observed bolometric flux and hence $d_L = (1+z) D_C$.

(b) Use the transverse size of a ruler to derive $d_A = D_C/(1+z)$.

(c) Combine the results to obtain distance duality.

(d) Which step in (a) would fail if photons disappeared into an unobserved channel? Does spatial flatness alone guarantee photon conservation?

Worked solution →

[3] Problem 15 — The Einstein–de Sitter toolkit. *Adapted from Yale ASTR 610, Problem Set 1, Problem 1.*

A flat universe contains pressureless matter only, with no radiation or Λ at any epoch in this idealized model.

(a) Derive $a/a_0 = (3H_0 t/2)^{2/3}$.

(b) Show that $\bar{\rho}(t) = 1/(6\pi G t^2)$ and $Ht = 2/3$.

(c) Derive the proper particle horizon.

(d) Evaluate the age and horizon for $H_0 = 70$, and explain why the horizon exceeds ct_0 without superluminal local light propagation.

Worked solution →

[3] Problem 16 — Can tired light pass every test? *Adapted from Yale ASTR 610, Problem 3, with additional observational checks.*

A static-space toy model proposes that photon energy obeys $dE/dr = -K\,E$ along its path.

(a) Solve for $E(r)$ and obtain $z(r)$.

(b) Show that the model reproduces a linear low-redshift Hubble law, and find K in Mpc^{-1} if that law has $H_0 = 70$.

(c) The model merely reduces each photon's energy and does not alter emission or arrival clock rates. What duration would it predict for a 20-day rest-frame supernova feature at $z = 1$? What does expansion predict?

(d) Explain why merely shifting a blackbody's photons in energy is not generally sufficient to preserve its observed temperature and intensity normalization. What additional assumptions would a competing model need?

Worked solution →

[3] Problem 17 — Equality in a different thermal universe. *Extension adapted from Yale ASTR 610, Problem 2.*

A hypothetical universe has $\Omega_\Lambda = 0$, a photon background at $T_0 = 10.0$ K, and a neutrino background that remains relativistic with energy density 1.5 times the photon energy density. Write $H_0 = 100h\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$.

(a) Show that $1 + z_{\mathrm{eq}} = \Omega_{m0}/(\Omega_{\gamma 0} + \Omega_{\nu 0})$.

(b) Calculate $\Omega_{\gamma 0}h^2$ for this temperature.

(c) Express $1 + z_{\mathrm{eq}}$ as a numerical coefficient times $\Omega_{m0}h^2$.

(d) Why is it essential that the neutrinos remain relativistic in this hypothetical model?

Worked solution →

[3] Problem 18 — Why flatness needs an explanation. *Extension adapted from MIT 8.286, Quiz 3 Review, Problem 9.*

For a universe with nonzero k, define Ω_{tot} excluding the curvature contribution.

(a) Show that $\Omega_{\mathrm{tot}} - 1 = kc^2/(a^2H^2)$ in the convention of Section 4.

(b) In an exactly matter-dominated era, find how $(\Omega_{\mathrm{tot}} - 1)/\Omega_{\mathrm{tot}}$ scales with t.

(c) Repeat for an exactly radiation-dominated era.

(d) During inflation with nearly constant H, how does $|\Omega_{\mathrm{tot}} - 1|$ scale with the number of e-folds? Explain why this helps the flatness problem without proving that k has changed.

Worked solution →

[3] Problem 19 — A very distant quasar: check the meaning of "distance." *Extension adapted from MIT 8.286, Quiz 3 Review, Problem 10. The historical source parameters are retained as a calculation, not a current record claim.*

Use $z = 5.82$, $H_0 = 65\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\Omega_{m0} = 0.35$, $\Omega_{\Lambda 0} = 0.65$, and neglect radiation.

(a) Write definite integrals for the present age, the age at emission, the present proper distance and the emission-time proper distance.

(b) Write the present proper-distance recession rate in terms of one of those distances.

(c) Evaluate the distances and times to two significant figures; numerical integration or the interactive distance lab is allowed.

(d) Explain how a present distance of roughly 27 billion light-years is compatible with a light-travel time below 14 billion years.

Worked solution →

[2] Problem 20 — A BAAO redshift calculation, with its approximation exposed. *Adapted from BAAO 2015 Round 2, Question 12.*

A galaxy has $z = 0.30$ and a rest-frame hydrogen line at $486.1\ \mathrm{nm}$. Take $H_0 = 72\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$. For parts (a)–(c), deliberately use the low-redshift approximation $v = cz = H_0 D$.

(a) Find the observed wavelength.

(b) Find the inferred distance in Mpc.

(c) Find the corresponding light-travel time in Gyr.

(d) Now use the flat $(0.3, 0.7)$ model with the same H_0 to estimate the lookback time and luminosity distance. Why do the answers differ?

(e) Which use of the relativistic Doppler formula would still not give the model's present proper-distance recession rate?

Worked solution →

19. Original Olympiad challenges, inspired by IOAA materials

[3] Problem 21 — A rocket against the Hubble flow. *MAIS original — inspired by IOAA 2022 Theory 4, "Journey Between Galaxies." New distances, speed choices and limiting-case analysis.*

A destination galaxy is initially $D_0 = 10.0\ \mathrm{Mpc}$ away and comoving with a background for which $H = 70\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ remains constant. A rocket leaves us toward it with constant **peculiar speed** u relative to the local comoving frame. Its proper separation $D(t)$ from us obeys

$$\frac{dD}{dt} = HD - u. \tag{63}$$

Ignore gravity, acceleration time and fuel limits.

(a) Solve for $D(t)$.

(b) Find the threshold launch speed for eventual arrival. Can a $1000\ \mathrm{km\,s^{-1}}$ rocket arrive?

(c) Find the travel time for $u = 1500\ \mathrm{km\,s^{-1}}$.

(d) Describe the limiting behaviour as u approaches the threshold from above.

(e) Why is D_0/u not the correct general answer? State one reason the constant-H assumption is not a full model of our universe.

Worked solution →

[3] Problem 22 — Weighing dark energy, then asking when it wins. *MAIS original — inspired by IOAA 2023 Theory 5, "Dark Energy." Adds a pressure and acceleration diagnostic.*

Assume spatial flatness, $H_0 = 70$, and present matter density $\rho_{m0} = 2.8 \times 10^{-27}\ \mathrm{kg\,m^{-3}}$. Neglect radiation for this late-time exercise. The remaining density is a cosmological constant.

(a) Find the dark-energy mass-equivalent density and energy density.

(b) Find Ω_{m0} and $\Omega_{\Lambda 0}$.

(c) Find the redshift of matter–Λ equality and the redshift at which acceleration begins.

(d) Find the present deceleration parameter.

(e) A student argues that negative pressure produces an inward pressure-gradient force, so it cannot accelerate expansion. Identify the conceptual error.

Worked solution →

[3] Problem 23 — A warm spectrum and a deliberately wrong clock. *MAIS original — inspired by IOAA 2024 Theory 5, "CMB." The matter-only assumption is supplied explicitly and then tested.*

An idealized observation identifies a CMB spectrum whose B_λ peak is at $0.500\ \mathrm{mm}$. Use $T_0 = 2.725\ \mathrm{K}$ and $b_\mathrm{W} = 2.898 \times 10^{-3}\ \mathrm{m\,K}$.

(a) Find its temperature, redshift and scale factor relative to today.

(b) In a flat matter-only model with $H_0 = 70$ and no other components at any time, estimate the cosmic age at that redshift.

(c) Explain why this is not a precision estimate for the real universe at that epoch.

(d) If the same wavelength shift were instead a local special-relativistic Doppler shift, calculate the corresponding speed. Why is this not the preferred interpretation of the CMB redshift?

Worked solution →

[3] Problem 24 — A small scar on a blackbody. *MAIS original — inspired by IOAA 2020 Team 14, "Spectral Distortions of the CMB." This is a simplified fixed-photon-number energy-budget model, not a thermalization calculation.*

Suppose an episode injects energy into the CMB after processes that change photon number have become ineffective. Assume the radiation subsequently relaxes to a Bose–Einstein spectrum

$$n(x) = \frac{1}{e^{x+\mu} - 1}, \qquad x = \frac{h\nu}{k_\mathrm{B} T}. \tag{64}$$

For small positive μ, the fractional changes relative to a blackbody at the same temperature are given as

$$\frac{\Delta N}{N} \simeq -1.368\mu, \qquad \frac{\Delta u}{u} \simeq -1.111\mu. \tag{65}$$

Photon number is fixed, so the temperature is allowed to adjust. Blackbody scalings are $N \propto T^3$ and $u \propto T^4$.

(a) Show that an injected fractional energy $\epsilon = \Delta u_\mathrm{inj}/u_i$ produces $\mu \simeq 1.40\epsilon$ to first order.

(b) Estimate μ for $\epsilon = 10^{-5}$ and the fractional temperature adjustment.

(c) Why would it be inconsistent to keep both photon number and temperature fixed after a positive energy injection?

(d) Explain why this estimate cannot be applied unchanged to an early epoch when photon-producing reactions remain efficient.

Worked solution →

[3] Problem 25 — Counting neutrons into helium. *MAIS original — inspired by the BBN topic of IOAA 2025 Theory 10. The assumptions below define a standalone neutron-counting exercise.*

At a stage just before efficient nucleosynthesis, the neutron-to-proton ratio is $r_0 = 1/6$. For the next 200 s, assume no weak interconversion, a free-neutron mean lifetime $\tau_n = 880$ s, and negligible expansion of the comoving volume. Then assume all surviving neutrons are incorporated into helium-4, with the remaining protons becoming hydrogen.

(a) Find the neutron-to-proton ratio at the end of the delay.

(b) Find the helium-4 mass fraction Y_p, neglecting small nuclear mass defects.

(c) Find the number ratio $n_{\mathrm{He}}/n_{\mathrm{H}}$.

(d) Why is the helium mass fraction not $r/(1+r)$?

(e) Explain qualitatively how a higher baryon density could change the delay and hence the helium abundance. Why is the real calculation more involved than this model?

Worked solution →

[3] Problem 26 — A hidden magnitude error is a hidden Hubble constant. *MAIS original — IOAA-style data and error reasoning, informed by the distance-ladder themes of IOAA 2023–2024 materials.*

A low-redshift supernova sample has accurately measured relative distances, but every assumed absolute magnitude is too bright by 0.050 mag: $M_{\mathrm{assumed}} - M_{\mathrm{true}} = -0.050$.

(a) Does the analysis overestimate or underestimate luminosity distances? Find the fractional bias.

(b) Find the corresponding fractional bias in H_0.

(c) If the biased estimate is 70.0, estimate the calibration-corrected value.

(d) Explain why the shape of a relative supernova Hubble diagram can still constrain acceleration even with an unknown constant magnitude offset.

(e) Would a magnitude error that grows with redshift be equally harmless? Explain.

Worked solution →

[3] Problem 27 — One ruler, two expansion measurements. *MAIS original — IOAA-style standard-ruler inference; connects CMB/BAO concepts rather than copying an official question.*

At $z = 0.50$, a galaxy survey measures a transverse BAO angle $\theta = 0.0300\ \text{rad}$ and a radial BAO redshift separation $\Delta z = 0.0450$. The comoving ruler is $r_d = 147\ \text{Mpc}$. Assume small angles and intervals, and a flat geometry.

(a) Find $D_M(z)$ from the transverse signal.

(b) Find $H(z)$ from the radial signal.

(c) Find d_A and d_L at this redshift.

(d) If the assumed r_d were 2% too large, how would the inferred D_M and H change?

(e) Explain why these measurements alone do not uniquely identify the microscopic nature of dark energy.

Worked solution →

[3] Problem 28 — Designing a curvature experiment. *MAIS original — IOAA-style geometric estimation, extending the website-sourced curved-space problem.*

A positively curved spatial geometry has curvature radius $R_c = 20.0\ \text{Gpc}$. Ideal observers measure a circle at geodesic radius $r = 2.00\ \text{Gpc}$ on one cosmic-time slice. Ignore expansion during the measurement and all observational systematics.

(a) Estimate the circumference deficit relative to $2\pi r$ using the first non-Euclidean term.

(b) Express the fractional deficit.

(c) If the circumference can be measured to 1% fractional precision, is this a convincing detection?

(d) Suggest two reasons a real cosmic curvature test is harder than this idealized circle.

(e) Why does failure to detect curvature not prove that the entire universe is spatially infinite?

Worked solution →

[3] Problem 29 — The last message that can arrive. *MAIS original — IOAA-style horizon reasoning, inspired by the expanding-background ideas in IOAA 2022 Theory 4.*

Assume the future universe is exactly de Sitter with constant $H = 70\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$. Two galaxies are comoving and initially separated by D_0.

(a) A light signal is sent toward us now. Solve for its proper distance $D(t)$ from us.

(b) Find the maximum D_0 for which it ever arrives, and express the answer in Gpc and Gly.

(c) For $D_0 = 0.8c/H$, find the arrival time.

(d) What happens at exactly $D_0 = c/H$?

(e) Distinguish this event horizon from the particle horizon and from the maximum distance of objects whose old light we see today.

Worked solution →

[3] Problem 30 — How much inflation is enough? *MAIS original — inspired by early-universe themes across IOAA materials and the Sheffield/MIT inflation exercises. A simplified, self-contained estimate.*

During inflation, H is approximately constant and the scale factor grows by e^N. Immediately afterward, suppose the universe is radiation-dominated until a matter-dominated era; for this estimate, use the supplied net result that the comoving Hubble radius grows by a factor 10^{26} between the end of inflation and today. Assume inflation starts with $|\Omega_{\mathrm{tot}} - 1| \sim 1$ and that the pre-inflationary region is causally connected on the initial Hubble scale.

(a) By what factor does $|\Omega_{\mathrm{tot}} - 1|$ decrease during N e-folds?

(b) Find the minimum N for it to be below 10^{-5} immediately after inflation, before the later growth.

(c) Including the supplied later growth factor, find the minimum N for a present deviation below 10^{-5}.

(d) Explain why the two bounds are so different, and why neither is a model-independent measurement of the actual number of e-folds.

(e) Explain how the same shrinking comoving Hubble radius addresses the horizon problem. Why does inflation not predict a perfectly smooth universe with no density perturbations?

Worked solution →

20. After the problems: explain what you learned

For each challenge, write a three-sentence explanation without equations: what changes, what stays fixed, and which observation would test the claim. Then compare your explanation with the **Idea** and **Checks** paragraphs in the Solutions. The aim is not to recognize a formula quickly; it is to know why that formula is the right description.

Download the practice data: synthetic Hubble catalogue (CSV). The catalogue's generating value is $H_0 = 72$ with fixed, non-random perturbations added for teaching. It must not be cited as an observational measurement of the real Hubble constant.